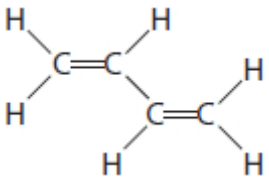
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Alkenes part A - MCQ

Q1.

The structure of a diene is shown.



How many σ bonds and π bonds are there in one molecule of this diene?

	σ bonds	π bonds
☒ **A**	7	2
☒ **B**	7	4
☒ **C**	9	2
☒ **D**	9	4

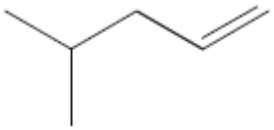
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(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the name of the product when this molecule reacts with chlorine gas?



- ☒ **A** 5,5-dichloro-2-methylpentane
- ☒ **B** 4,5-dichloro-2-methylpentane
- ☒ **C** 2,3-dichloro-4-methylpentane
- ☒ **D** 1,2-dichloro-4-methylpentane

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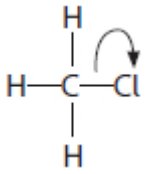
(Total for question = 1 mark)

Q3.

Curly arrows are used in reaction mechanisms.

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The curly arrow shown represents the movement of

(1)

- ☒ **A** an electron from a bond to an atom, forming free radicals
- ☒ **B** an electron from a bond to an atom, forming ions
- ☒ **C** a pair of electrons from a bond to an atom, forming free radicals
- ☒ **D** a pair of electrons from a bond to an atom, forming ions

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(Total for question = 1 mark)

Q4.

What is the systematic name of compound **X**?



Compound **X**

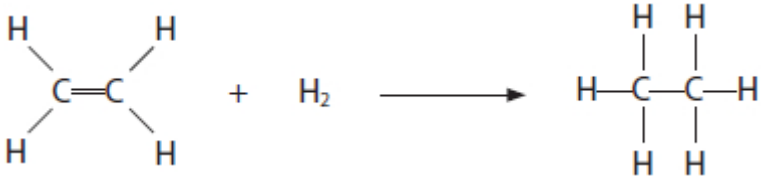
- ☒ **A** *E*-2-chlorobut-2-ene
- ☒ **B** *Z*-2-chlorobut-2-ene
- ☒ **C** *E*-3-chlorobut-2-ene
- ☒ **D** *Z*-3-chlorobut-2-ene

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(Total for question = 1 mark)

Q5.

Ethene reacts with hydrogen in the presence of a heated nickel catalyst to form ethane.



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Which types of bond are broken and formed in this reaction?

	Bonds broken	Bonds formed
☒ A	σ only	π only
☒ B	π only	σ only
☒ C	σ and π	σ only
☒ D	σ and π	π only

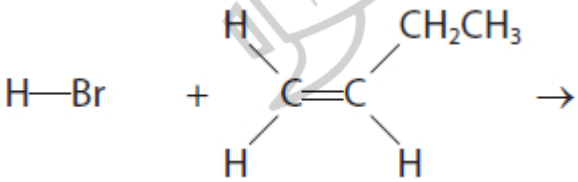
(Total for question = 1 mark)

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Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the major product of this reaction?

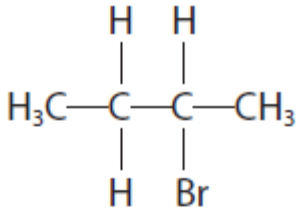


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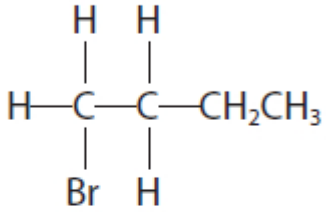
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☒ **A**

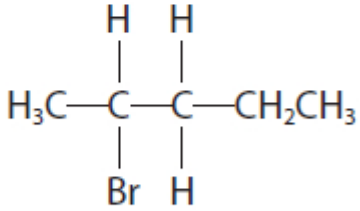


☒ **B**

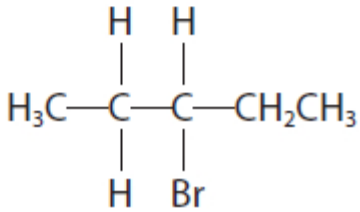


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☒ **C**



☒ **D**



(Total for question = 1 mark)

Q7.

During a cracking reaction, each molecule of an alkane with formula $C_{10}H_{22}$ formed only two molecules of ethene and one molecule of hydrocarbon **A**.

What is the molecular formula of **A**?

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(1)

☒ **A** C_6H_{10}
☒ **B** C_6H_{14}
☒ **C** C_8H_{16}
☒ **D** C_8H_{18}

(Total for question = 1 mark)

Q8.

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When hydrogen bromide, HBr, reacts with propene, a higher percentage of 2-bromopropane forms than 1-bromopropane.

Which is the best explanation for this?

(1)

- ☒ **A** 1-bromopropane is more stable than 2-bromopropane
- ☒ **B** 2-bromopropane is more stable than 1-bromopropane
- ☒ **C** a primary carbocation is more stable than a secondary carbocation
- ☒ **D** a secondary carbocation is more stable than a primary carbocation

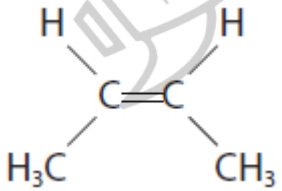
(Total for question = 1 mark)

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Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

But-2-ene shows geometric isomerism.



What are the prefixes that could be used in naming this isomer?

	cis / trans	*E / Z*
☒ **A**	*cis*	*E*
☒ **B**	*cis*	*Z*
☒ **C**	*trans*	*E*
☒ **D**	*trans*	*Z*

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(Total for question = 1 mark)

Q10.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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When zinc is added to copper(II) sulfate solution, copper is formed.

(a) Which of these is the best name for this type of reaction?

(1)

- ☒ **A** addition
- ☒ **B** displacement
- ☒ **C** neutralisation
- ☒ **D** substitution

(b) Which is the ionic half-equation for a process that takes place during this reaction?

(1)

- ☒ **A** $Cu^{2+} + e^- \rightarrow Cu^+$
- ☒ **B** $Cu^+ + e^- \rightarrow Cu$
- ☒ **C** $Zn \rightarrow Zn^+ + e^-$
- ☒ **D** $Zn \rightarrow Zn^{2+} + 2e^-$

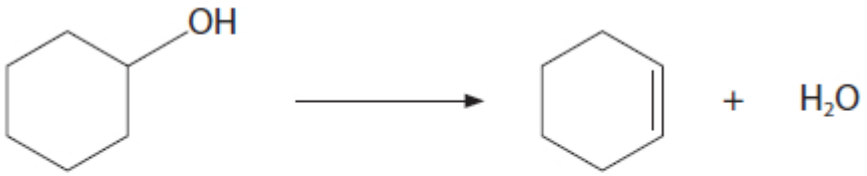
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(Total for question = 2 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Cyclohexene may be prepared by the dehydration of cyclohexanol.



cyclohexanol
$M_r = 100$

cyclohexene
$M_r = 82$

What mass of cyclohexene can be made from 12.5 g of cyclohexanol if the yield is 51.2 %?

- ☒ **A** 5.25 g
- ☒ **B** 6.40 g
- ☒ **C** 7.80 g
- ☒ **D** 10.25 g

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(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these is **not** a way of limiting global problems caused by polymer disposal?

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- ☒ **A** developing biodegradable polymers
- ☒ **B** exporting polymer waste
- ☒ **C** removing toxic waste gases produced by the incineration of polymers
- ☒ **D** reusing products made from polymers

(Total for question = 1 mark)

Q13.

Geometric isomerism is shown by 2-chlorobut-2-ene.

(a) What is the skeletal formula of *E*-2-chlorobut-2-ene?

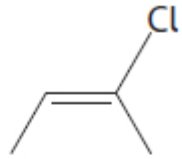
(1)

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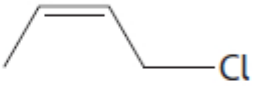
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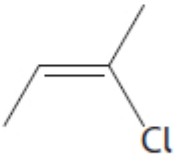
☒ **A**



☒ **B**

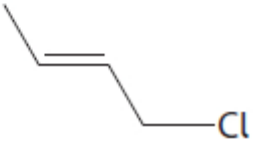


☒ **C**



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☒ **D**



(b) What is the **total** number of sigma bonds in *Z*-2-chlorobut-2-ene?

(1)

☒ **A** 3
☒ **B** 4
☒ **C** 10
☒ **D** 11

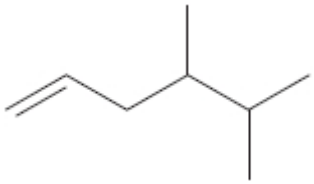
(Total for question = 2 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the systematic name of this compound?

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☒ **A** 1,1,2-trimethylpent-4-ene
☒ **B** 2,3-dimethylhex-5-ene
☒ **C** 4,5-dimethylhex-1-ene
☒ **D** 4,5,5-trimethylpent-1-ene

(Total for question = 1 mark)

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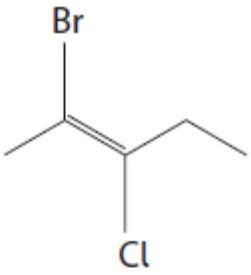
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Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

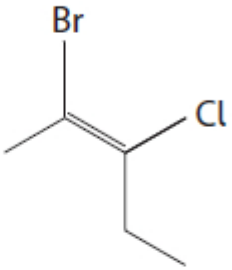
Which structure has the name *E*-2-bromo-3-chloropent-2-ene?

☐ **A**



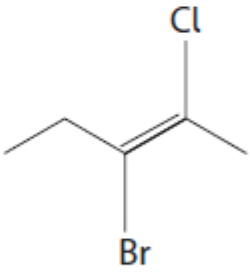
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☐ **B**

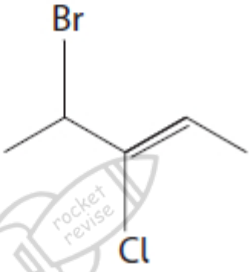


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☐ **C**



☐ **D**



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(Total for question = 1 mark)

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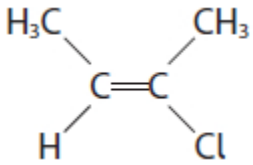
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Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the IUPAC name for the compound with the structure shown?



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- ☒ **A** *cis*-2-chlorobut-2-ene
- ☒ **B** *trans*-2-chlorobut-2-ene
- ☒ **C** *E*-2-chlorobut-2-ene
- ☒ **D** *Z*-2-chlorobut-2-ene

(Total for question = 1 mark)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A hydrocarbon **X** has a molar mass of 98 g mol^{-1}. When a sample of **X** is shaken with bromine water, the colour of the bromine water does **not** change.

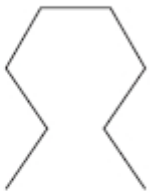
Which of these could be the structure of **X**?

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☒ **A** 

☒ **B** 

☒ **C** 

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☒ **D** 

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(Total for question = 1 mark)

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Q18.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Propene, $CH_3CH{=}CH_2$, reacts with hydrogen chloride, HCl, to form halogenoalkanes.

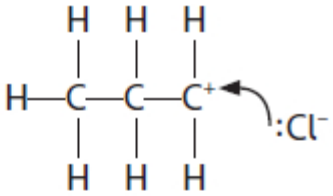
(a) Which of these steps is most likely to occur in the reaction?

(1)

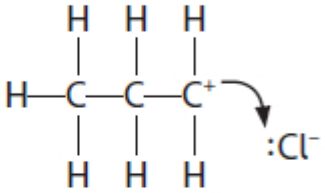
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☒ **A**

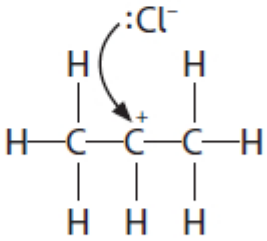


☒ **B**

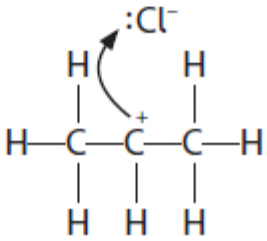


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☒ **C**



☒ **D**



(b) The reaction of propene with hydrogen chloride is an example of

(1)

☒ **A** free radical substitution
☒ **B** free radical addition
☒ **C** electrophilic substitution
☒ **D** electrophilic addition

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(Total for question = 2 marks)

Q19.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A cyclic hydrocarbon has the structure shown.

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How many pi (π) and sigma (σ) bonds are present in the molecule?

	pi (π) bonds	sigma (σ) bonds
☒ **A**	2	11
☒ **B**	2	5
☒ **C**	4	3
☒ **D**	4	10

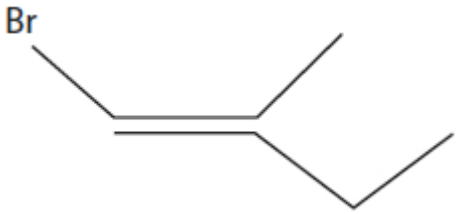
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(Total for question = 1 mark)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the IUPAC name for the compound shown?



☒ **A** *E*-1-bromo-2-methylbut-1-ene
☒ **B** *Z*-1-bromo-2-methylbut-1-ene
☒ **C** *E*-1-bromo-2-ethyl-2-methylethene
☒ **D** *Z*-1-bromo-2-ethylpropene

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(Total for question = 1 mark)

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Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which molecule can exist as *E-Z* isomers?

- ☐ **A** 2-chloro-4-ethylhex-3-ene
- ☐ **B** 3-chloro-2-methylhex-2-ene
- ☐ **C** 2,3-dimethylhex-2-ene
- ☐ **D** 2,5-dichlorohex-3-ene

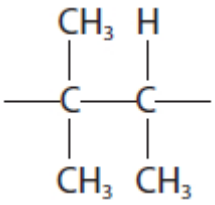
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(Total for question = 1 mark)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The repeat unit of a polymer is shown.



Which is the name of the monomer that forms this polymer?

- ☐ **A** 1,1,2-trimethylethene
- ☐ **B** 1,1-dimethylpropene
- ☐ **C** 2-methylbut-2-ene
- ☐ **D** 3-methylbut-2-ene

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(Total for question = 1 mark)

Q23.

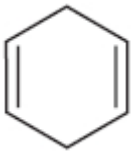
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The skeletal formula of cyclohexa−1,4−diene is shown.



(a) The total number of σ (sigma) bonds in this molecule is

(1)

- ☐ **A** 4
- ☐ **B** 6
- ☐ **C** 12
- ☐ **D** 14

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(b) 1.60 g of cyclohexa−1,4−diene is reduced to cyclohexane by reacting it with hydrogen gas.

[The molar volume of a gas is 24.0 dm^3 mol^{-1} at room temperature and pressure (r.t.p.)
M_r of cyclohexa−1,4−diene = 80]
What is the minimum volume, in dm^3, of hydrogen needed at r.t.p.?

(1)

- ☐ **A** 0.0400
- ☐ **B** 0.480
- ☐ **C** 0.960
- ☐ **D** 1.92

(Total for question = 2 marks)

Q24.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Ethene reacts with acidified potassium manganate(VII). A student makes four statements about the reaction.

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- the organic product can show geometric isomerism
- the type of reaction that takes place is oxidation
- the bond angles around each carbon atom are greater in the product than in ethene
- the acidified potassium manganate(VII) decolourises in the reaction

How many of these statements are correct?

- ☐ **A** one
- ☐ **B** two
- ☐ **C** three

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☒ **D** four

(Total for question = 1 mark)

Q25.

Which of these is not a chemical reaction?

(1)

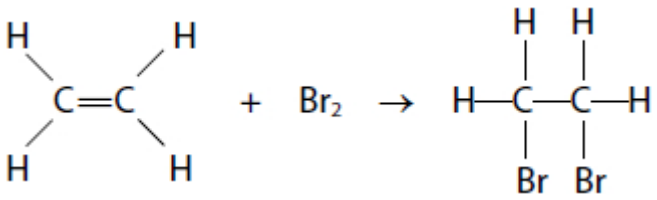
☒ **A** cracking
☒ **B** fractional distillation
☒ **C** polymerisation
☒ **D** reforming

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(Total for question = 1 mark)

Q26.

Ethene reacts with bromine to form 1,2-dibromoethane.



For the ethene molecule, what is the type of bond broken and the type of bond fission occurring in this reaction?

	Bond broken	Bond fission
☒ **A**	π	heterolytic
☒ **B**	π	homolytic
☒ **C**	σ	heterolytic
☒ **D**	σ	homolytic

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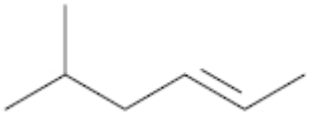
(Total for question = 1 mark)

Q27.

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What is the systematic name for this compound?



(1)

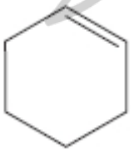
- ☒ **A** *E*-5-methylhex-2-ene
- ☒ **B** *Z*-5-methylhex-2-ene
- ☒ **C** *E*-2-methylpent-4-ene
- ☒ **D** *Z*-2-methylpent-4-ene

(Total for question = 1 mark)

Q28.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

This question is about cyclohexene, a cyclic alkene.



(a) What is the general formula of cyclic alkenes such as cyclohexene?

(1)

- ☒ **A** C_nH_{2n-4}
- ☒ **B** C_nH_{2n-2}
- ☒ **C** C_nH_{2n}
- ☒ **D** C_nH_{2n+2}

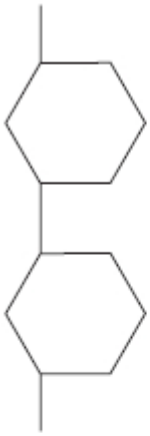
(b) Cyclohexene can form an addition polymer.

Which diagram shows two repeat units of this addition polymer?

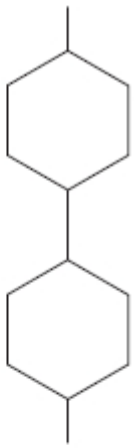
(1)

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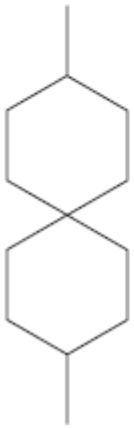


☒ **A**

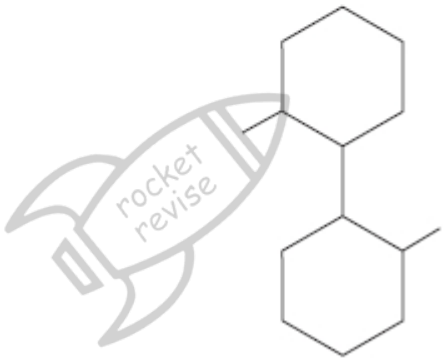


☒ **B**

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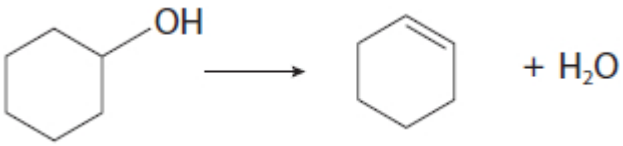


☒ **C**



☒ **D**

(c) Cyclohexanol can be converted into cyclohexene.



What mass of cyclohexanol is needed to make 7.20 g of cyclohexene, if the yield of this reaction is 72.0%?
[M_r values: cyclohexanol = 100 cyclohexene = 82.0]

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(1)

☒ **A** 4.25 g
☒ **B** 5.90 g
☒ **C** 8.78 g
☒ **D** 12.2 g

(Total for question = 3 marks)

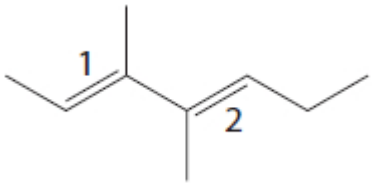
Q29.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The molecule shown has two double bonds.



Which is the correct identification for each double bond?

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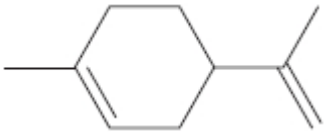
	Double bond 1	Double bond 2
☐ **A**	*E*	*E*
☐ **B**	*E*	*Z*
☐ **C**	*Z*	*E*
☐ **D**	*Z*	*Z*

(Total for question = 1 mark)

Q30.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about limonene.



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(a) What is the molecular formula of limonene?

(1)

☐ **A** $C_{10}H_{10}$
☐ **B** $C_{10}H_{16}$
☐ **C** $C_{10}H_{18}$
☐ **D** $C_{10}H_{22}$

(b) Limonene is completely hydrogenated by hydrogen in the presence of a nickel catalyst. What is the **empirical formula** of the product formed?

(1)

☐ **A** C_5H_{11}

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☒ **B** C_5H_9
☒ **C** C_5H_8
☒ **D** CH_2

(Total for question = 2 marks)

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which statement about poly(ethene) is **not** correct?

☒ **A** it is an addition polymer
☒ **B** it decolourises bromine water
☒ **C** it is non-biodegradable
☒ **D** it has the empirical formula CH_2

(Total for question = 1 mark)

Q32.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about the production of ethanol by fermentation of glucose or by hydration of ethene.

(a) What is the percentage atom economy by mass for the production by fermentation?

$$C_6H_{12}O_6 \rightarrow 2CH_3CH_2OH + 2CO_2$$

(1)

☒ **A** 24.4%
☒ **B** 25.6%
☒ **C** 48.9%
☒ **D** 51.1%

(b) What is the percentage atom economy by mass for the production by hydration?

$$C_2H_4 + H_2O \rightarrow CH_3CH_2OH$$

(1)

☒ **A** 39.1%
☒ **B** 60.9%
☒ **C** 78.2%
☒ **D** 100%

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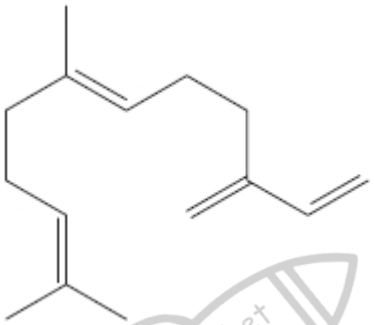
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(Total for question = 2 marks)

Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

β-Farnesene is an alarm pheromone released by aphids.



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How many π bonds and how many σ bonds are present in this molecule?

- ☒ **A** 4π and 31σ
- ☒ **B** 4π and 38σ
- ☒ **C** 8π and 28σ
- ☒ **D** 8π and 34σ

(Total for question = 1 mark)

Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these formulae are correct for the compound 1,4-diethylcyclohexene?



(i)
skeletal formula

$C_{10}H_{18}$

(ii)
molecular formula

C_5H_9

(iii)
empirical formula

- ☒ **A** none of the three
- ☒ **B** (i) only
- ☒ **C** (ii) and (iii) only

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☒ **D** (i), (ii) and (iii)

(Total for question = 1 mark)

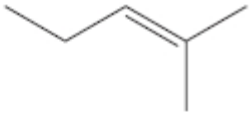
Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

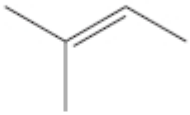
Which molecule has a geometric isomer?

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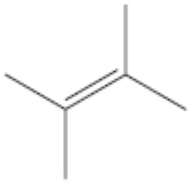
☒ **A**



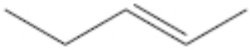
☒ **B**



☒ **C**



☒ **D**



(Total for question = 1 mark)

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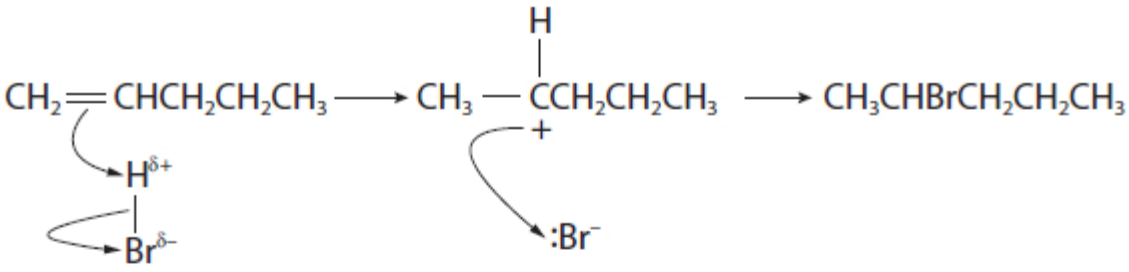
Q36.

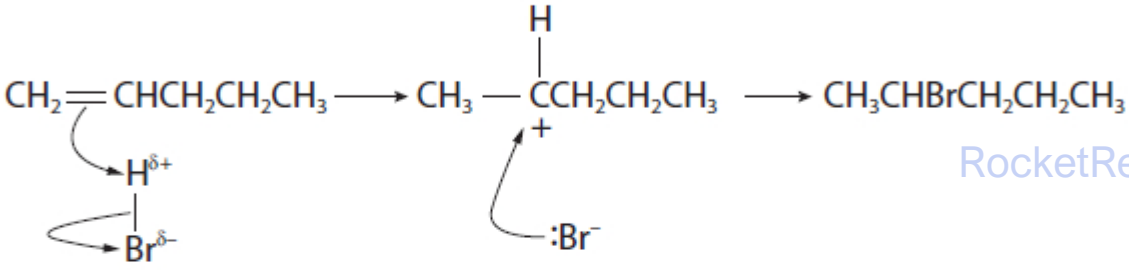
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the correct mechanism for the electrophilic addition reaction between HBr and pent-1-ene to give the major product?

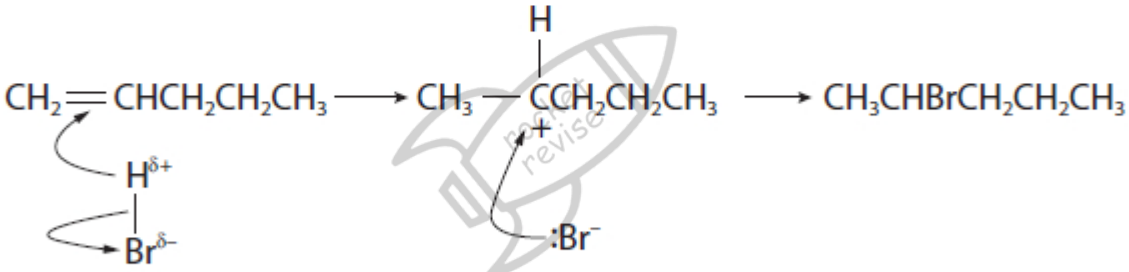
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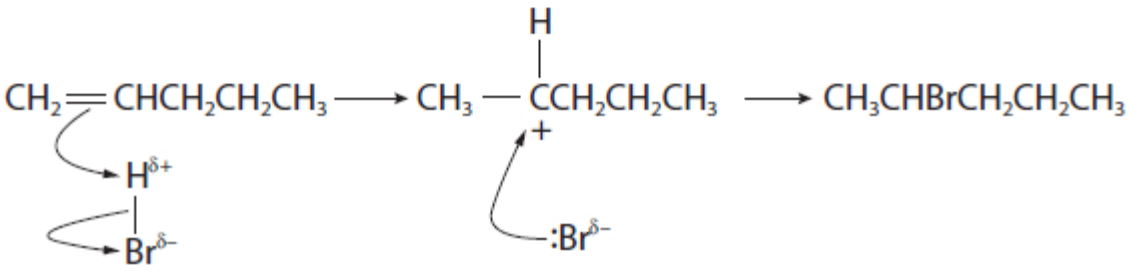
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☒ **A** 

☒ **B** 

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☒ **C** 

☒ **D** 

(Total for question = 1 mark)

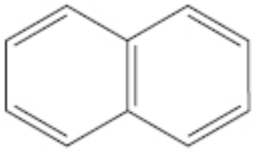
Q37.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

What is the molecular formula of naphthalene?

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☒ **A** $C_{10}H_8$
☒ **B** $C_{10}H_{10}$
☒ **C** $C_{12}H_{10}$
☒ **D** $C_{12}H_{12}$

(Total for question = 1 mark)

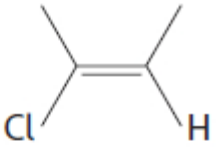
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Q38.

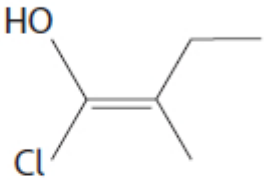
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which structure is a *Z*-isomer?

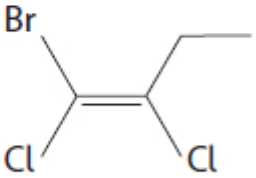
☒ **A**



☒ **B**

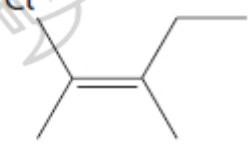


☒ **C**



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☒ **D**



(Total for question = 1 mark)

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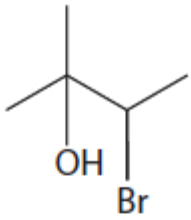
Q39.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Bromine and bromine water can be used to test for a carbon-carbon double bond.
When **bromine water** reacts with 2-methylbut-2-ene, there is one major organic product.

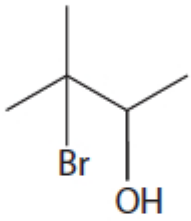
Which is the skeletal formula of the **major** product?

☐ **A**

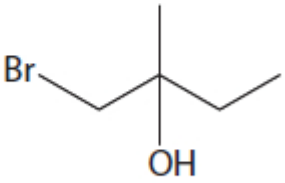


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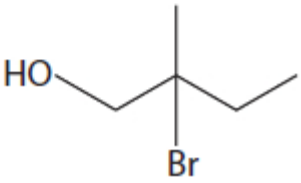
☐ **B**



☐ **C**



☐ **D**



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(Total for question = 1 mark)

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Alkenes part B - Long Questions

Q1.

This question is mainly about alkenes.

(a) A few drops of bromine water are added to separate test tubes of propane and propene and the mixtures are shaken.

Describe what is seen at the end of each experiment.

(2)

Propane

.....

Propene

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.....

(b) Give the reagents and conditions for each of these conversions.

(i) $CH_2{=}{=}CH_2 \rightarrow CH_3CH_2OH$

(1)

.....

.....

(ii) $CH_2{=}{=}CH_2 \rightarrow CH_2OHCH_2OH$

(1)

.....

.....

(c) Draw the structure of *Z*-3-methylpent-2-ene.

(1)

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(d) Exactly 720 cm^3 of hydrogen gas, measured at room temperature and pressure (r.t.p.), reacted with 0.010 mol of an alkene to form an alkane.

Deduce the number of double bonds in one molecule of the alkene.
You **must** show your working.
[Molar volume of gas at r.t.p. = 24 000 $cm^3\ mol^{-1}$]

(2)

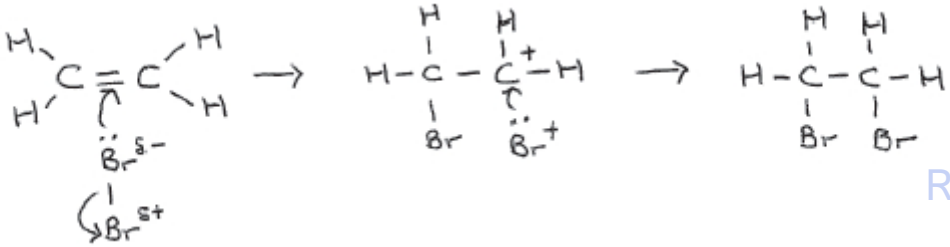
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(e) A student drew a mechanism for the addition of bromine to ethene.



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Describe the three changes needed to correct this student's mechanism.

(3)

Change 1

.....

.....

.....

Change 2

.....

.....

.....

Change 3

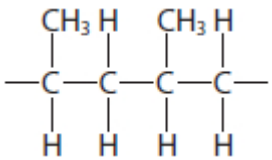
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(f) Part of the structure of a polymer is shown.



Draw the structure of the monomer used to make this polymer.

(1)

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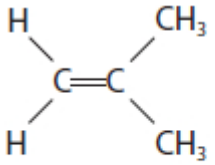
(Total for question = 11 marks)

Q2.

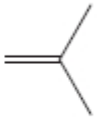
This question is about the alkene 2-methylpropene.

The formulae show two different ways of representing a molecule of 2-methylpropene.

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formula **1**



formula **2**

(a) Give the **empirical** formula of 2-methylpropene.

(1)

.....

(b) Give a reason why 2-methylpropene does **not** show geometric isomerism.

(1)

.....

.....

.....

.....

(c) Draw the mechanism for the reaction between 2-methylpropene and bromine, Br_2. Include curly arrows, and relevant lone pairs and dipoles.

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Use formula 1 to represent 2-methylpropene.

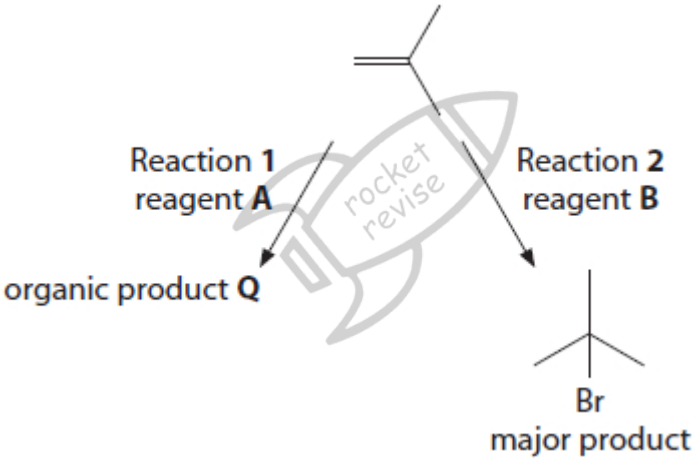
(4)

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(d) Two reactions of 2-methylpropene are shown.



(i) In Reaction **1** the reagent **A** is acidified potassium manganate(VII).
Give the **skeletal** formula of organic product **Q**.

(1)

(ii) Give the colour change seen during Reaction **1**.

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(1)

From ..... to .....

(iii) Identify, by name or formula, reagent **B** in Reaction **2**.

(1)

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(iv) Explain why 2-bromo-2-methylpropane is the major organic product in Reaction **2**.

(2)

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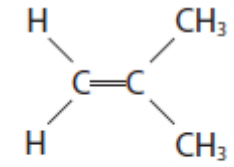
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(e) Draw **two** repeat units of poly(2-methylpropene).



2-methylpropene

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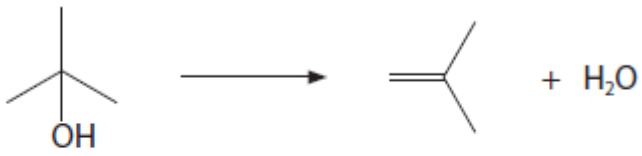
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(f) A sample of 2-methylpropene was prepared from 2-methylpropan-2-ol.



2-methylpropan-2-ol
$M_r = 74.0$

2-methylpropene
$M_r = 56.0$

The yield of this reaction was 58.2%.
Calculate the mass of 2-methylpropene formed from 6.85 g of 2-methylpropan-2-ol.
Give your answer to an appropriate number of significant figures.

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(4)

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(Total for question = 17 marks)

Q3.

This question is about fuels and polymers.

Used coffee grounds have been suggested as a carbon-neutral fuel to replace some fossil fuels.

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(a) (i) Explain why coffee grounds might be considered a carbon-neutral fuel.

(2)

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(ii) Explain how the use of fossil fuels causes climate change.

(2)

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(b) Long chain alkanes are not normally used as fuels as they produce soot.

(i) Name another pollutant formed by **incomplete** combustion of alkanes.

(1)

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(ii) Write the equation for the **complete** combustion of octane.
State symbols are not required.

(2)

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(c) Long chain alkanes are converted into smaller, more useful molecules including alkenes.

(i) Name this process.

(1)

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(ii) Give a test for alkenes, including the positive result.

(2)

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(d) Alkenes, such as ethene, can be used to make polymers.

(i) Write a balanced equation for the polymerisation of ethene using displayed formulae.

(1)

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(ii) Bananas produce ethene as they ripen.
Suggest one advantage and one disadvantage of using ripening bananas as a source of ethene for polymer production.

(2)

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(e) Burning poly(chloroethene) in an incinerator results in the formation of hydrogen chloride.

(i) State a **hazard** associated with hydrogen chloride.

(1)

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(ii) Suggest how the hydrogen chloride could be removed from the waste gases produced in an incinerator.

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(1)

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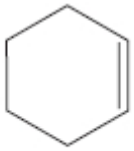
(Total for question = 15 marks)

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Q4.

This question is about the liquids cyclohexene, C_6H_{10}, and bromine, Br_2.



cyclohexene

(a) Calculate the volume, in cm^3, of 0.0300 mol of cyclohexene.

Give your answer to an appropriate number of significant figures.
You must show your working.

Data A_r values: H = 1.0 C = 12.0

Density of cyclohexene = 0.811 g cm^{-3}

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(2)

(b) Bromine reacts with cyclohexene.

(i) State the colour change when an excess of cyclohexene is added to liquid bromine.

(2)

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(ii) This reaction is very similar to the reaction of ethene with bromine.
Draw the mechanism for this reaction of cyclohexene with bromine.
Include curly arrows, and any relevant dipoles and lone pairs.

(4)

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(c) Bromine vapour in the atmosphere can cause irritation to the eyes at a concentration of 1.1 ppm and to the lungs at a concentration of 10 ppm.

(i) State the meaning of ppm.

(1)

.....

(ii) A sample of 3.25 cm^3 of bromine vapour leaks into a laboratory.
Calculate the minimum volume, in **dm^3**, of the laboratory for the resulting concentration of bromine to be at a safe level.

(3)

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(d) Because bromine is so hazardous it is usually used in the laboratory in aqueous solution.

A sample of bromine reacts with exactly 0.0300 mol of cyclohexene.
Calculate the minimum volume of water needed to completely dissolve this sample of bromine at room temperature.

Data A_r value: Br = 79.9

Concentration of bromine in water at room temperature = 35 g dm^{-3}

(3)

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(Total for question = 15 marks)

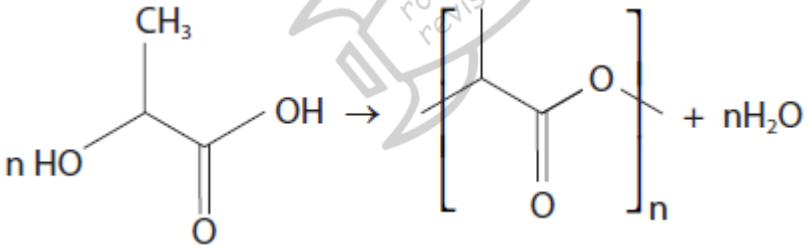
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Q5.

This question is about polymers.

(a) Plastic bags can be made from poly(lactic acid) (PLA), which is biodegradable.

Lactic acid is obtained from corn.
An equation for the polymerisation of lactic acid is shown.



(i) Write a balanced equation for the polymerisation of propene using displayed formulae.

(2)

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(ii) Give **one** difference between the polymerisation of propene and that of lactic acid.

(1)

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(iii) State the meaning of the term biodegradable.

(1)

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(iv) Give **three** advantages of biodegradable polymers.

(3)

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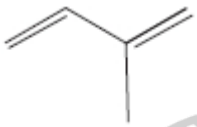
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(b) Isoprene is used to make synthetic rubber. The skeletal formula for isoprene is shown.



(i) State why isoprene does **not** have geometric isomers.

(1)

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(ii) A related molecule does show geometric isomerism.

Draw the geometric isomer of this molecule.

(1)

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(iii) Give a reason why a double bond may result in geometric isomerism.

(1)

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(Total for question = 10 marks)

Q6.

This question is about alkenes.

(a) An alkene has a molar mass of 112 g mol^{-1}.

Deduce the molecular formula of this alkene.

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(1)

(b) There are a number of different alkenes with the molecular formula C_4H_8.

(i) Draw the structure of the **branched-chain** alkene with the molecular formula C_4H_8.

(1)

(ii) Give the structures and names of the two geometric isomers with the molecular formula C_4H_8.

(2)

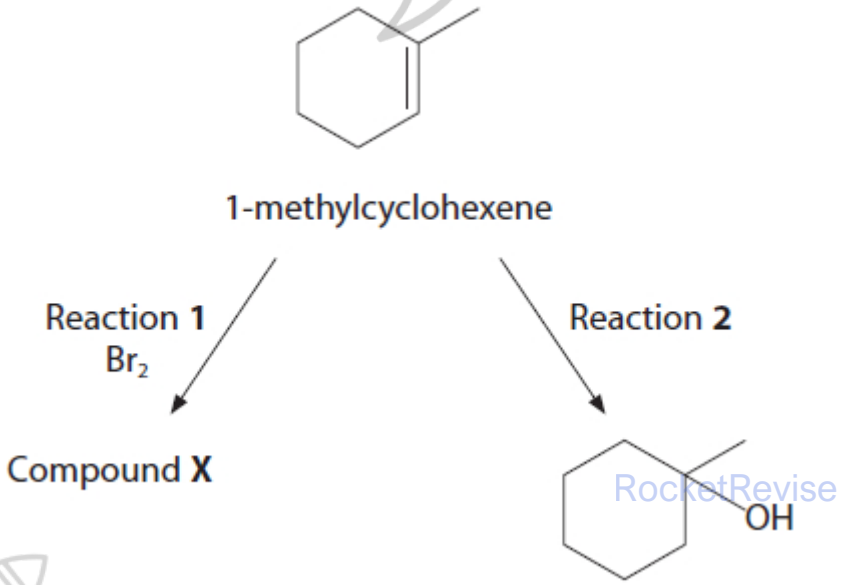
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Structure of geometric isomer 1	Structure of geometric isomer 2
Name of isomer 1	Name of isomer 2

(c) Two reactions of 1-methylcyclohexene are shown.



(i) Draw the **skeletal** formula of compound **X** formed in Reaction **1**.

(1)

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(ii) Give the reagent and condition needed for Reaction **2**.

(2)

Reagent .....

Condition .....

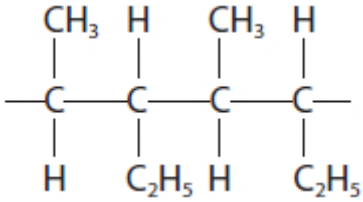
(d) Iodine monochloride, ICl, reacts with alkenes in a similar way to hydrogen bromide.

Complete the mechanism for the reaction of iodine monochloride with propene to form the **major** product.
Include curly arrows, the relevant lone pair and the structures of the intermediate and product.

(4)



(e) A section of a polymer showing two repeat units is given.



Give the **name** of the monomer that forms this polymer.

(1)

.....

(f) 0.0100 mol of an alkene reacts completely with exactly 600 cm^3 of hydrogen gas, measured at 298 K and 1.24×10^5 Pa
pressure, to form an alkane.

Use the ideal gas equation to deduce the number of double bonds in **one** molecule of the alkene.
You **must** show your working.
[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(4)

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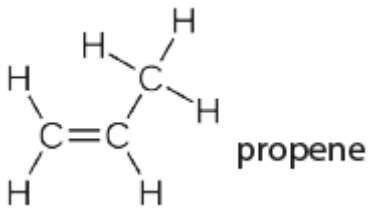
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(Total for question = 16 marks)

Q7.

This question is about the reactions of propene.



propene

(a) Write an equation for the incomplete combustion of **one mole** of propene to form carbon dioxide, carbon monoxide,
carbon and water as the only products. Include state symbols.

(2)

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(b) State **one** similarity and **one** difference that would be **seen** when propene is mixed with separate samples of
acidified potassium manganate(VII) solution and of bromine water.

(2)

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(c) Propene reacts by addition polymerisation to form poly(propene).

Draw the structure of poly(propene), showing **two** repeat units.

(1)

(d) Propene reacts with bromine monochloride, BrCl, to form 1-bromo-2-chloropropane as the major product.

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(i) Complete the diagram of bromine monochloride to show the dipole.

(1)

Br ——— Cl

(ii) Draw the mechanism for the formation of 1-bromo-2-chloropropane in this reaction.
Include curly arrows, and relevant lone pairs.

(3)

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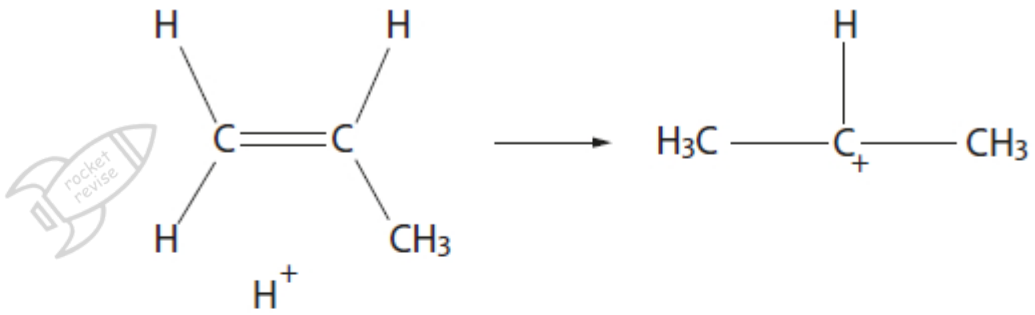
(e) Propene reacts with steam in the presence of an acid catalyst to form a mixture of the alcohols propan-1-ol and propan-2-ol.

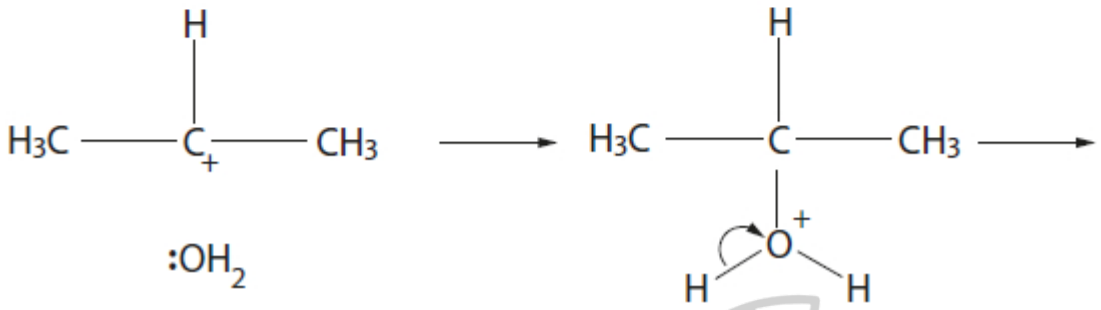
Complete the mechanism for the formation of propan-2-ol, by adding curly arrows.
Include the species formed in the final step.

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(3)



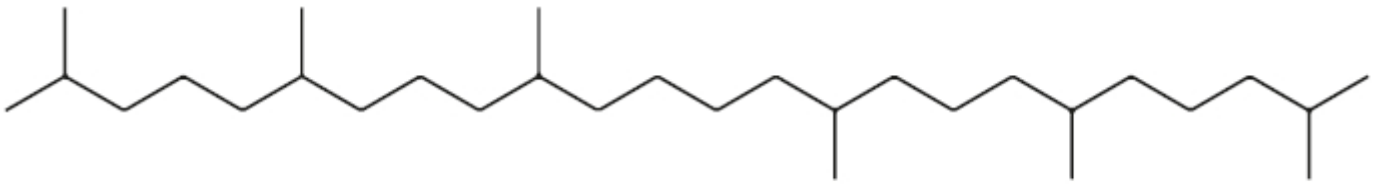


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(Total for question = 12 marks)

Q8.

This question is about the production of squalane, a liquid alkane which occurs naturally in human skin and is used in cosmetics.



squalane

(a) Suggest **two** properties that make squalane useful in cosmetics. RocketRevise

(2)

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(b) Give the **molecular** formula of squalane.

(1)

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(c) Squalane can be produced from squalene, an alkene present in shark liver oil, by reaction with hydrogen gas in the presence of a suitable catalyst.

(i) Give the name of a suitable catalyst for the hydrogenation of squalene.

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(1)

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(ii) Squalane used in cosmetic products must contain no more than 0.2 ppm by mass of catalyst.
Calculate the maximum permitted mass of catalyst in a product containing 50 g of squalane.

(1)

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(iii) A reactor at 200°C contains 8500 mol of liquid squalene, and hydrogen gas at a pressure of 4.0×10^5 Pa.
Under these conditions, the complete hydrogenation of squalene requires 500 m^3 of hydrogen gas.
Calculate the number of C=C bonds in one molecule of squalene.
You **must** show your working.
[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(4)

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(iv) Write the equation, using molecular formulae, for the complete hydrogenation of squalene to

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squalane.
State symbols are **not** required.

(1)

(d) Globally, 2.8 million dm3 of squalene is used each year.
Traditionally squalene was obtained exclusively from shark liver oil, which is a mixture of liquids.

The liver of a shark yields 300 g of squalene.
(i) Suggest the name of a suitable technique to obtain squalene from shark liver oil.

(1)

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(ii) Calculate the minimum number of sharks that would be needed to produce 2.8 million dm^3 of squalene.
[Density of squalene = 0.86 g cm^{-3}]

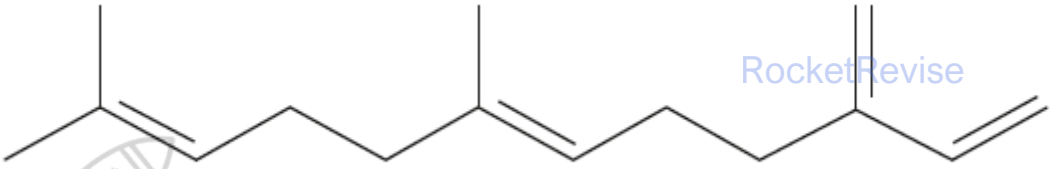
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(2)

(iii) Many large corporations now use squalane obtained entirely from plants.
Squalane can be obtained sustainably from corn starch with a yield of 23 % by mass.
The production of 1 tonne of corn starch requires 0.093 hectares of land.
Calculate the area of land, **in km^2**, required to produce 2500 tonnes of squalane from corn starch.
[1 tonne = 1000kg 1 hectare = 0.01 km^2]

(3)

(e) The *E*-isomer of beta-farnesene can also be obtained from corn starch.



E-beta-farnesene

(i) Explain why beta-farnesene exhibits geometric isomerism and has only two geometric isomers.
You may label the structure and use this in your answer.

(2)

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(ii) Draw the **skeletal** formula of the geometric isomer of *E*-beta-farnesene, giving a reason why this is named the
Z-isomer.

(2)

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(f) The compound alpha-farnesene, $C_{15}H_{24}$, is a structural isomer of beta-farnesene.

The structural formula of alpha-farnesene is

$$(CH_3)_2C=CHCH_2CH_2C(CH_3)=CHCH_2CH=C(CH_3)CH=CH_2$$

(i) State what is meant by the term **structural** isomers.

(2)

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(ii) State the number of **geometric** isomers of alpha-farnesene.

(1)

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(iii) Complete the diagram to show another structural isomer of $C_{15}H_{24}$.

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(1)



(Total for question = 24 marks)

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Q9.

Alkanes and alkenes are obtained from crude oil.

(a) Describe how a sample of octadecane can be obtained from a mixture of alkanes.

(2)

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(b) (i) Octadecane can be cracked to produce butene and **one** other product.

Complete the equation. State symbols are not required.

$$C_{18}H_{38} \rightarrow 2C_4H_8 +$$

(1)

(ii) One of the products of this cracking reaction is but-1-ene.

Give the **skeletal** formulae for the other three **alkene** isomers of C_4H_8

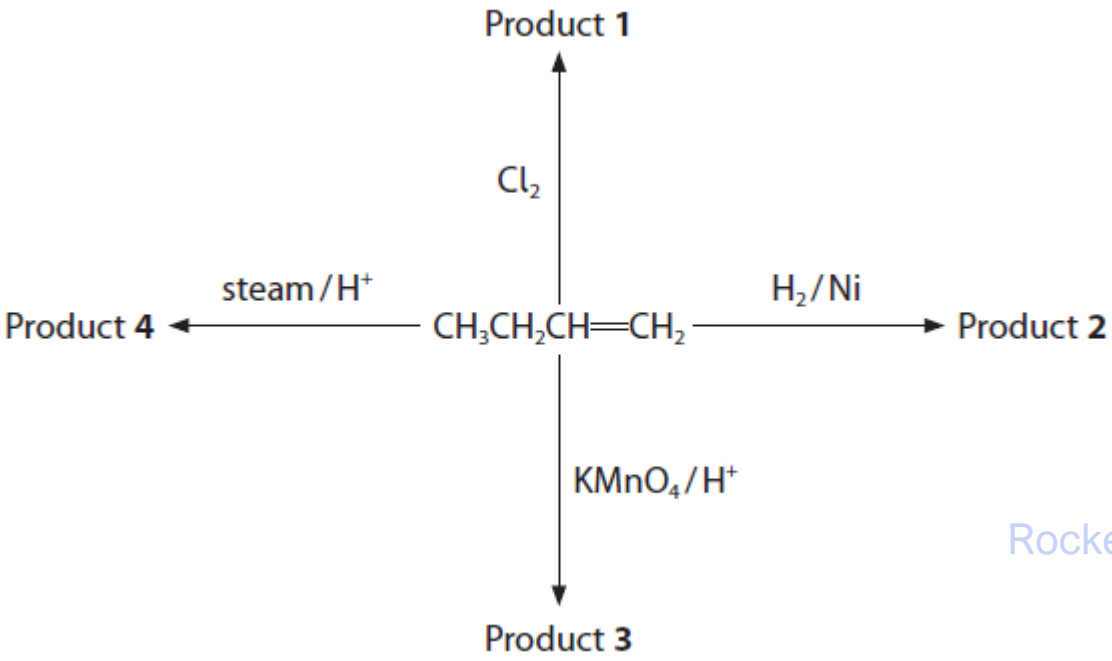
(2)

(iii) Some reactions of but-1-ene are shown.

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Give the name and **structural** formula of each of the products.

(4)

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Product	Name	Structural formula
1		
2		
3		
4		

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(c) Draw the displayed formula of poly(but-1-ene) showing two repeat units.

(2)

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(d) State **one** advantage and **one** disadvantage of using incineration for the disposal of polymers, other than the effect on climate.

(2)

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(e) (i) Butane is used as a fuel.

The equation for the complete combustion of butane is shown.

$$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$$

35.0 cm^3 butane is completely burned in 300 cm^3 oxygen.
Calculate the final total volume of gas in cm^3 .
All volumes are measured at the same temperature and pressure.

(3)

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(ii) Explain the main hazard when using butane as a fuel in a portable heater in an enclosed space.

(2)

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(Total for question = 18 marks)

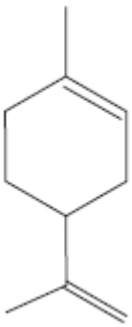
Q10.

Limonene is an oil extracted from the skin of citrus fruits such as oranges or lemons.
It is found in some cleaning products such as those used to remove chewing gum.

Limonene has the structure shown.

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(a) Deduce the molecular formula of limonene.

(2)

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(b) (i) A few drops of acidified potassium manganate(VII) solution are mixed with a sample of limonene.

State the colour change.

(1)

From ..... to .....

(ii) Draw the skeletal formula of the structure of the **organic** product formed when the reaction in (b)(i) is carried out with **excess** acidified potassium manganate(VII).

(1)

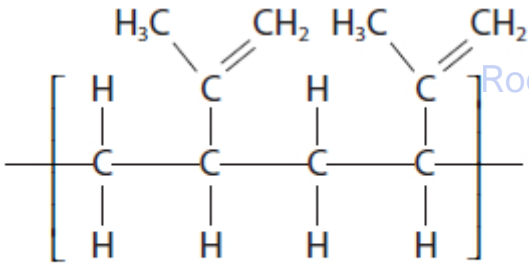
(iii) Name the type of reaction that limonene undergoes in (b)(i).

(1)

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(c) On heating, limonene reacts to form two molecules of a smaller hydrocarbon, **Q**.

Q can be polymerised to form **R**, a polymer used to improve the grip of car tyres.
A section of **R** is shown.



Section of polymer **R**

Draw the structure of **Q**.

(1)

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(d) Give **two** ways in which chemists can help minimise problems caused by the disposal of polymers.

(2)

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(e) Limonene is extracted from orange peel using a solvent.

Possible solvents are hexane and liquid carbon dioxide.
Liquid carbon dioxide can be obtained by allowing solid carbon dioxide to melt under high pressure.

(i) Suggest one advantage and one disadvantage of using liquid carbon dioxide rather than hexane to dissolve the limonene. Assume that the carbon dioxide does not escape into the atmosphere.

(2)

Advantage

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.....

Disadvantage

.....

.....

(ii) Orange peel contains 1.63 % of limonene by mass.
A chewing gum remover contains 30 cm^3 of limonene per bottle.
Calculate the mass of orange peel, in kg, needed to obtain 30 cm^3 of limonene.
[Density of limonene = 0.841 g cm^{-3}]

(3)

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(Total for question = 13 marks)

Q11.

Organic waste may be disposed of by landfill or incineration.
Both processes produce gases.

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(a) The main gases produced from a typical landfill are shown in the table.

Gas	Percentage by volume / %
methane	50
carbon dioxide	45
nitrogen	4
sulfur compounds	1

(i) Name the process that forms these gases in landfill.

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(1)

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(ii) State the **main** environmental problem caused by landfill gases, identifying the gas or gases responsible.

(2)

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(iii) One tonne of landfill waste produces approximately 12.5 dm^3 of landfill gases per day.
Calculate the mass of carbon dioxide produced in a year by a typical landfill site which contains 90 000 tonnes of waste.
Assume that the gas volume is measured at room temperature and pressure (r.t.p.).
[Molar volume of gas at r.t.p. = 24.0 dm^3 mol^{-1}]

(3)

(b) Suggest **two** advantages of incineration over landfill.

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(2)

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(c) Environmental groups prefer recycling to both landfill and incineration.

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Suggest **one** advantage of recycling.

(1)

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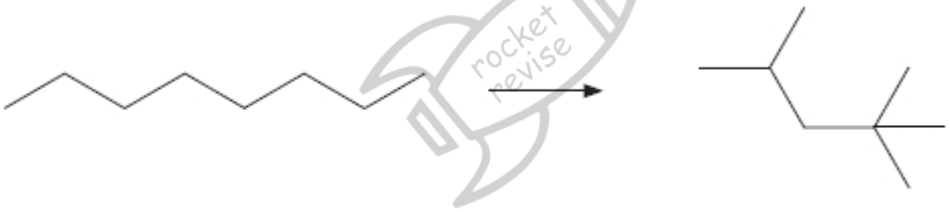
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Q12.

This question is about hydrocarbons obtained from crude oil.

(a) Petrol used in cars has a high proportion of the alkane iso-octane. Iso-octane can be formed by heating octane to 400 °C in the presence of a catalyst.



octane iso-octane

(i) Give the IUPAC name for iso-octane.

(1)

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(ii) Name the industrial process used to convert octane into iso-octane.

(1)

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(iii) Give a reason why petrol needs a high proportion of iso-octane.

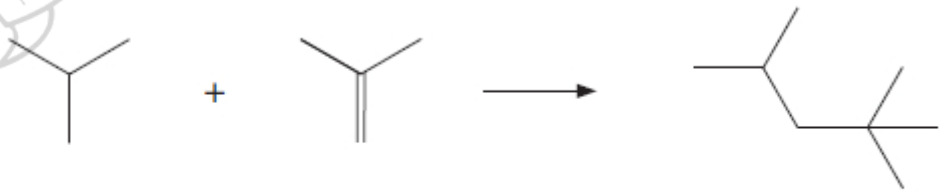
(1)

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(b) Another way to produce iso-octane is by reacting iso-butane with 2-methylpropene.



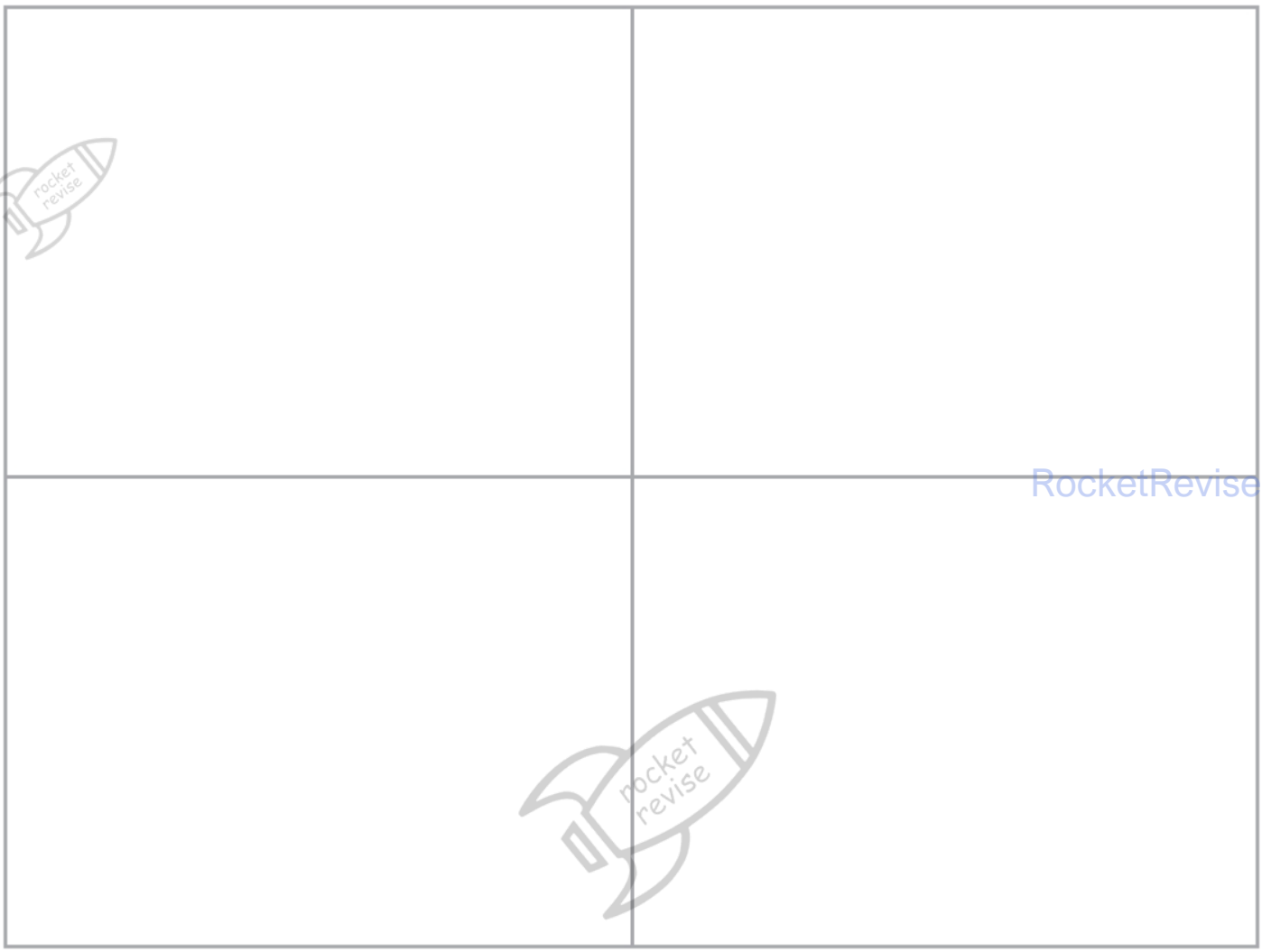
iso-butane 2-methylpropene iso-octane

Draw the **skeletal** formulae of four isomers of 2-methylpropene.
Include only **one** isomer that does not decolorise bromine water.

(4)

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(c) 2-methylpropene reacts with hydrogen bromide to form two possible products.

(i) Draw the mechanism for the formation of the **major** product of the reaction between 2-methylpropene and hydrogen bromide, HBr.
Include curly arrows, and relevant lone pairs and dipoles.

(4)

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(ii) Give the reason why your product in (c)(i) is the major product.

(1)

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(d) Another component of petrol, compound **X**, contains 92.3 % carbon and 7.7 % hydrogen by mass.

(i) Calculate the empirical formula of compound **X**.
You must show all your working.

(2)

(ii) A sample of compound **X** has a mass of 0.267 g and vaporises at 85.0 °C and 104 kPa to produce a gaseous sample with a volume of 98.0 cm^3.
Calculate the molar mass of compound **X**.
You must show all your working.
[$pV = nRTR$ = 8.31 J mol^{-1} K^{-1}]

(4)

(iii) Deduce the molecular formula of compound **X**, using your answers to (d)(i) and (d)(ii).

(1)

(Total for question = 19 marks)

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Q13.

The elements carbon and silicon both form dioxides.

(a) Carbon dioxide is a simple covalent molecule but silicon dioxide has a giant covalent structure.

(i) Describe the covalent bond between a silicon atom and an oxygen atom in silicon dioxide, in terms of the particles involved.

(2)

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(ii) Compare and contrast the covalent bonding in carbon dioxide and silicon dioxide in terms of orbital overlap.

(3)

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(b) The shape of the carbon dioxide molecule affects its physical properties.

(i) Explain the shape of the carbon dioxide molecule.

(3)

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(ii) Explain the polarity of the carbon–oxygen bond.

(2)

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(iii) State whether or not the carbon dioxide molecule is polar.
Justify your answer.

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(1)

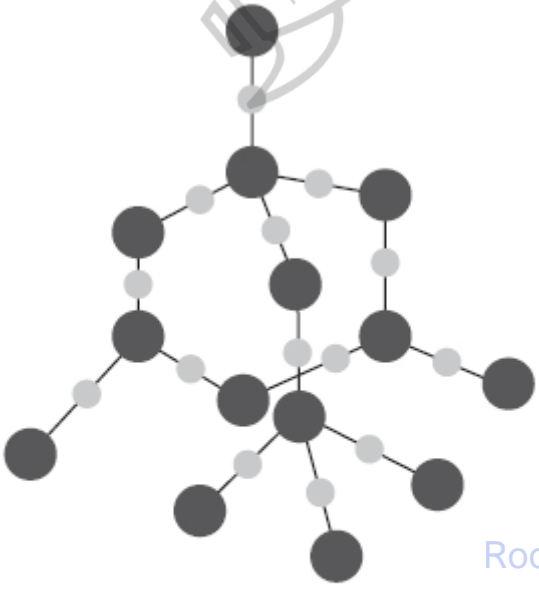
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(c) The structure of silicon dioxide may be referred to as a diamond structure.



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silicon

oxygen

(i) Using your knowledge of the structure of diamond, suggest how the structure of silicon dioxide is similar to that of diamond.

(1)

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(ii) Give a possible reason why silicon dioxide has a lower melting temperature than diamond, even though the Si—O bond is stronger than the C—C bond.

(1)

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(Total for question = 13 marks)

Q14.

This question is about alkenes.

Alkenes contain a carbon to carbon double bond that consists of a σ bond and a π bond.

(a) Complete the diagram to show the areas of electron density for each bond. Label the σ bond and the π bond.

(2)

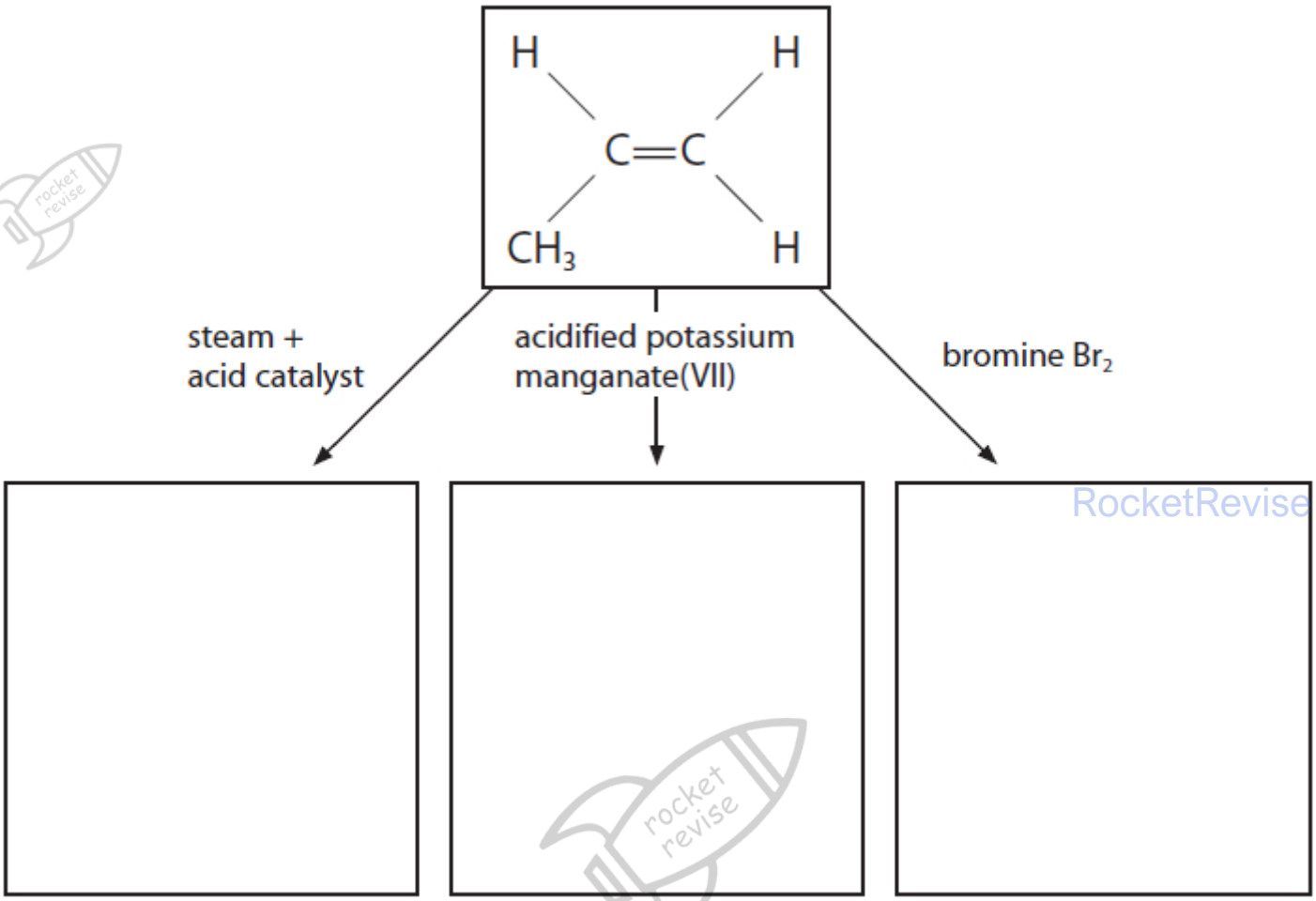
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(b) Propene, C_3H_6, is an alkene. The reagents needed for three reactions of propene are shown.

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(i) In each box, draw the structure of the organic product of the reaction.

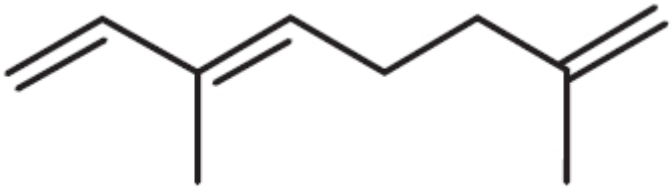
(3)

(ii) Propene also reacts with hydrogen bromide, HBr.
Give the mechanism for this reaction to form the **major** product.
Include curly arrows, and relevant lone pairs and dipoles.

(3)

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(c) Alpha-ocimene contains three carbon to carbon double bonds. It is found in plants and has a sweet smell. The skeletal formula of alpha-ocimene is shown.



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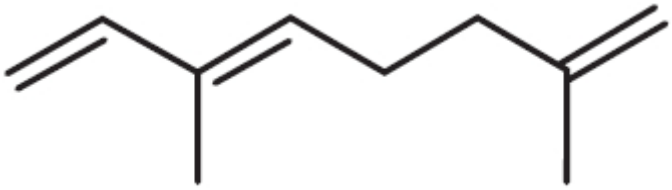
(i) Give the molecular formula of alpha-ocimene.

(1)

.....

(ii) On the skeletal formula, draw a circle around the part of the molecule that gives rise to the geometric isomerism of alpha-ocimene.

(1)



(iii) Draw the **skeletal** formula of the other geometric isomer of alpha-ocimene.

(1)

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(iv) In an experiment, 0.050 mol of alpha-ocimene reacted with 3.6 dm^3 of hydrogen, H_2, in the presence of a catalyst.
Deduce the structure of the product of this reaction.
You **must** show your working.
[Molar volume of H_2 = 24 dm^3 mol^{-1}]

(3)

Calculation

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Structure

(Total for question = 14 marks)

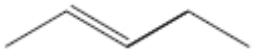
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Q15.

This question is about alkenes.

(a) A straight-chain alkene, **X**, has the structure shown.



(i) Draw the skeletal formulae of the four structural isomers of **X** that are alkenes. **Do not** include geometric isomers.

(4)

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(ii) Explain why **X** can exist as **geometric** isomers.

(2)

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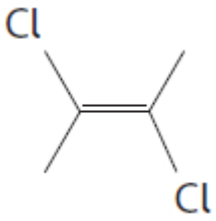
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(b) Another compound, **Y**, has the structure shown.



(i) Give the IUPAC name for **Y**.

(1)

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(ii) A student stated that
'Compound **Y** has polar bonds so it is a polar molecule'.
Comment on this statement.

(3)

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(c) Bisabolenes are a group of compounds found in plants and fungi. Bisabolene molecules contain several carbon–carbon double bonds.

5.51 g of α-bisabolene reacted with 1873 cm^3 of hydrogen gas, H_2.
The reaction occurred at a temperature of 150 °C and a pressure of 152 000 Pa.
[M_r value: α-bisabolene = 204.0]

(i) Calculate the amount, in mol, of α-bisabolene in 5.51 g.

(1)

(ii) Calculate the amount, in mol, of hydrogen gas that reacted with α-bisabolene. Use the ideal gas equation, $pV = nRT$.
[R = 8.31 J mol^{-1} K^{-1}]

(3)

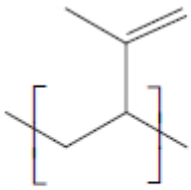
(iii) Deduce the number of carbon-carbon double bonds in a molecule of α-bisabolene, using your answers to (c)(i) and (c)(ii).

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(1)

(d) The repeat unit of the polymer polyisoprene is shown.



polyisoprene

RocketRevise

(i) Draw the **skeletal** formula of the monomer of polyisoprene.

(1)

(ii) A sample of polyisoprene has a relative molecular mass of 50 250.
Calculate the number of repeat units in the sample.

(2)

(e) Incinerators used to dispose of polymer waste have a system to remove waste gases.
Suggest why these systems contain solid calcium hydroxide and why it is finely powdered.

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(2)

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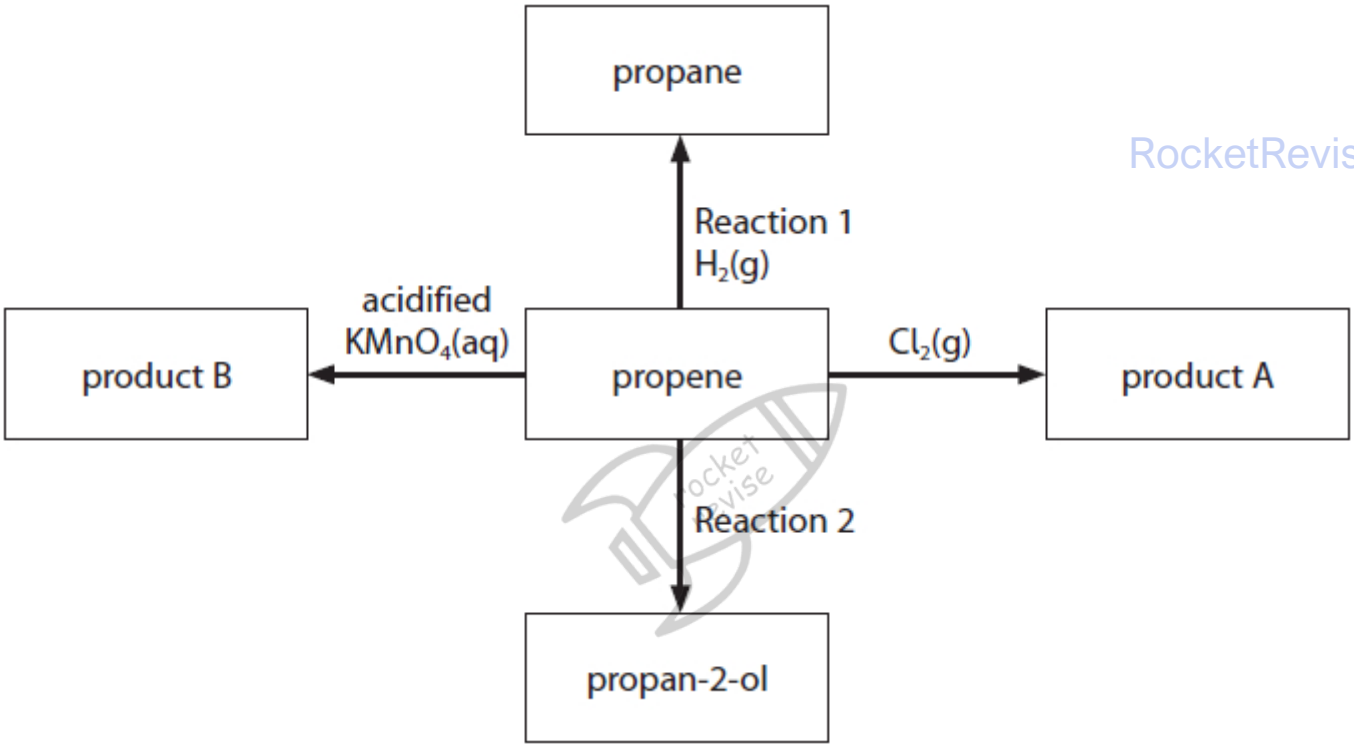
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(Total for question = 20 marks)

Q16.

This question is about propene and some related compounds.

(a) Some reactions of propene are shown.



(i) Identify, by name or formula, the catalyst needed for Reaction 1.

(1)

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(ii) Name product A.

(1)

.....

(iii) Identify, by name or formula, the substances needed for Reaction 2.

(2)

.....

.....

(iv) Draw the **displayed** formula of product B.

(1)

(b) (i) Propene can undergo addition polymerisation.

Write the equation for the polymerisation of propene.

(2)

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(ii) Over 70 million tonnes of poly(propene) are produced each year.
State **two** problems with the disposal of polymers such as poly(propene).

(2)

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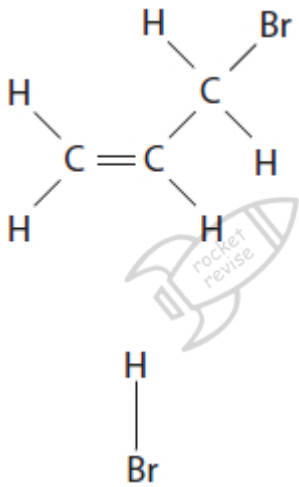
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(c) Hydrogen bromide reacts with 3-bromopropene to form 1,2-dibromopropane as the major product.

(i) Complete the mechanism to show this reaction.
Include curly arrows, and any relevant lone pairs and dipoles.

(4)



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(ii) Explain why 1,2-dibromopropane is the major product instead of 1,3-dibromopropane.

(2)

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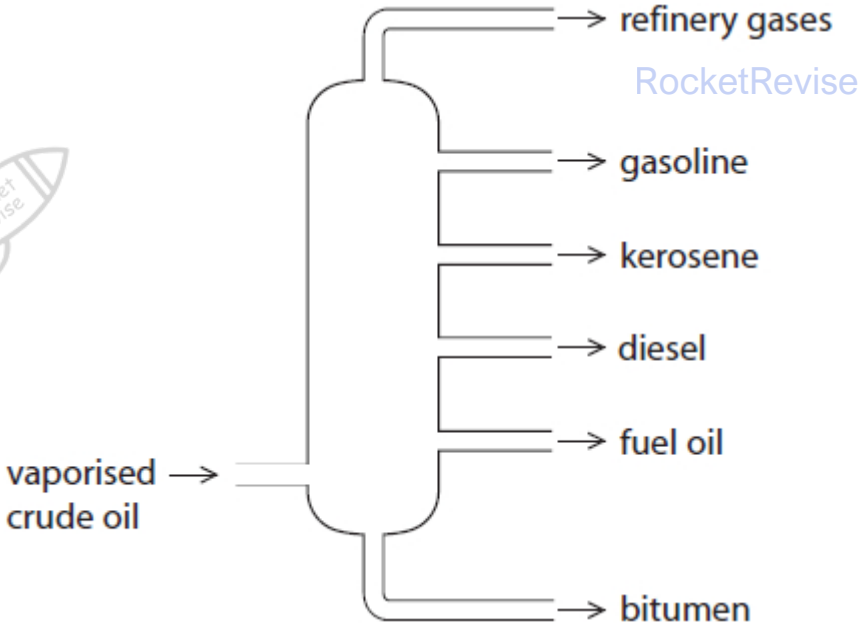
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(Total for question = 15 marks)

Q17.

Crude oil is mainly a mixture of saturated hydrocarbons that can be separated by fractional distillation. The diagram shows, in a simplified form, the products of fractional distillation.



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(a) (i) State what is meant by a saturated hydrocarbon.

(2)

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(ii) Describe how the fractions in crude oil are separated during fractional distillation.

(2)

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(b) Butane, C_4H_{10}, is found in the refinery gases fraction of crude oil.

It is used as a propellant in pharmaceutical inhalers.
An inhaler that provides a total of 120 doses contains 1.55 g of butane.
Calculate the volume, in cm^3, of propellant used for each dose at 25 °C and 100 kPa.
Use the ideal gas equation and give your answer to an appropriate number of significant figures.
[$pV = nRT$ $R = 8.31$ J K^{-1} mol^{-1}]

(5)

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(c) Octane, C_8H_{18}, is found in the gasoline fraction of crude oil.

(i) Write an equation for the complete combustion of octane. Include state symbols.

(2)

(ii) Give **two** reasons why alternative fuels, such as bioethanol, are being developed to replace those produced from crude oil.

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(2)

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(iii) Straight-chain alkanes, such as octane, are converted into branched-chain alkanes to improve the performance of petrol in car engines.
Name this conversion process.

RocketRevise

(1)

.....

(iv) Give an equation using skeletal formulae for the conversion of octane into 2,5-dimethylhexane.
State symbols are not required.

(1)

(d) In steam cracking, a mixture of hydrocarbons and steam is heated to 850 °C for a very short time in the absence of oxygen.

(i) Give a reason why oxygen should be excluded during steam cracking.

(1)

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(ii) Ethene is the major product.
State one use of ethene.

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(1)

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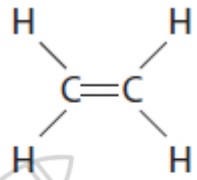
(e) Ethene reacts with hydrogen bromide to form bromoethane.

Draw the mechanism for this reaction.
Include curly arrows, and relevant lone pairs and dipoles.

(3)

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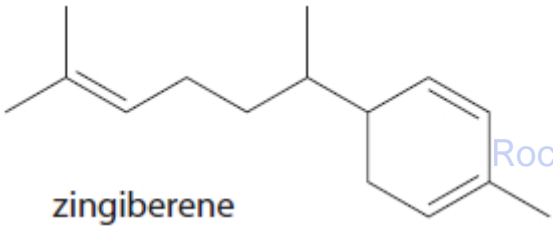
(Total for question = 20 marks)

Q18.

Zingiberene is the compound that gives ginger its characteristic flavour. Its IUPAC name is 2-methyl-5-(6-methylhept-5-en-2-yl)cyclohexa-1,3-diene.

(a) On the structure of zingiberene, draw a circle around the '2-methyl' group referred to in the IUPAC name.

(1)



zingiberene

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(b) Deduce the molecular formula of zingiberene.

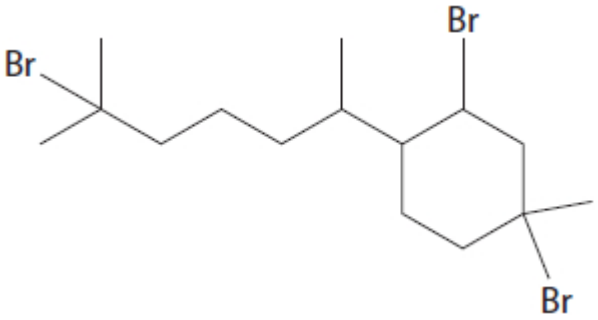
(2)

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(c) When zingiberene reacts with excess hydrogen bromide, there are a number of possible products. The structure of the major product is shown.

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(i) Name the type and mechanism of the reaction.

(1)

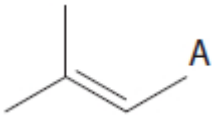
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(ii) The diagram shows a simplified structure of zingiberene, in which part of the molecule is represented by A.

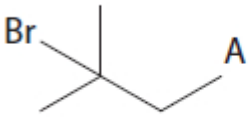
Complete the mechanism for the reaction of zingiberene with **one** molecule of hydrogen bromide.

Include curly arrows, and any relevant dipoles and lone pairs.

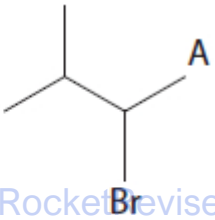
(4)



(iii) For the reaction in (c)(ii) there are two possible products:



I



II

Explain why I is the major product, by referring to your mechanism.

(2)

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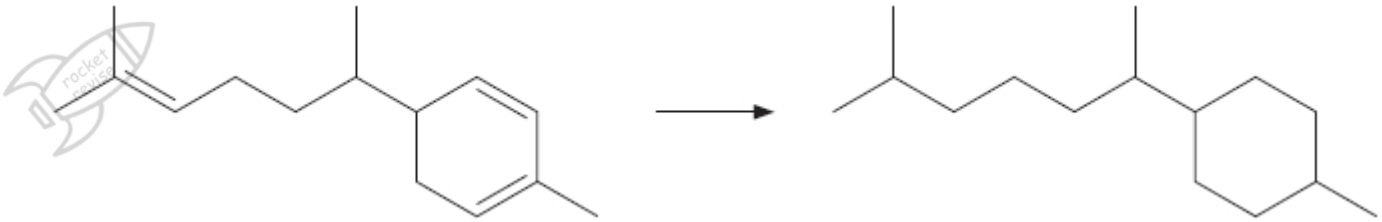
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(d) Zingiberene reacts with hydrogen gas in the presence of a catalyst.



(i) Identify the catalyst, by name or formula.

(1)

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(ii) 2.0 mol of zingiberene react completely with hydrogen at 150 °C and a pressure of 120 kPa. Calculate the minimum volume of hydrogen needed under these conditions, stating your units.
[Ideal gas equation is $pV = nRT$ Gas constant (R) = 8.31 J mol^{-1} K^{-1}]

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(4)

(Total for question = 15 marks)

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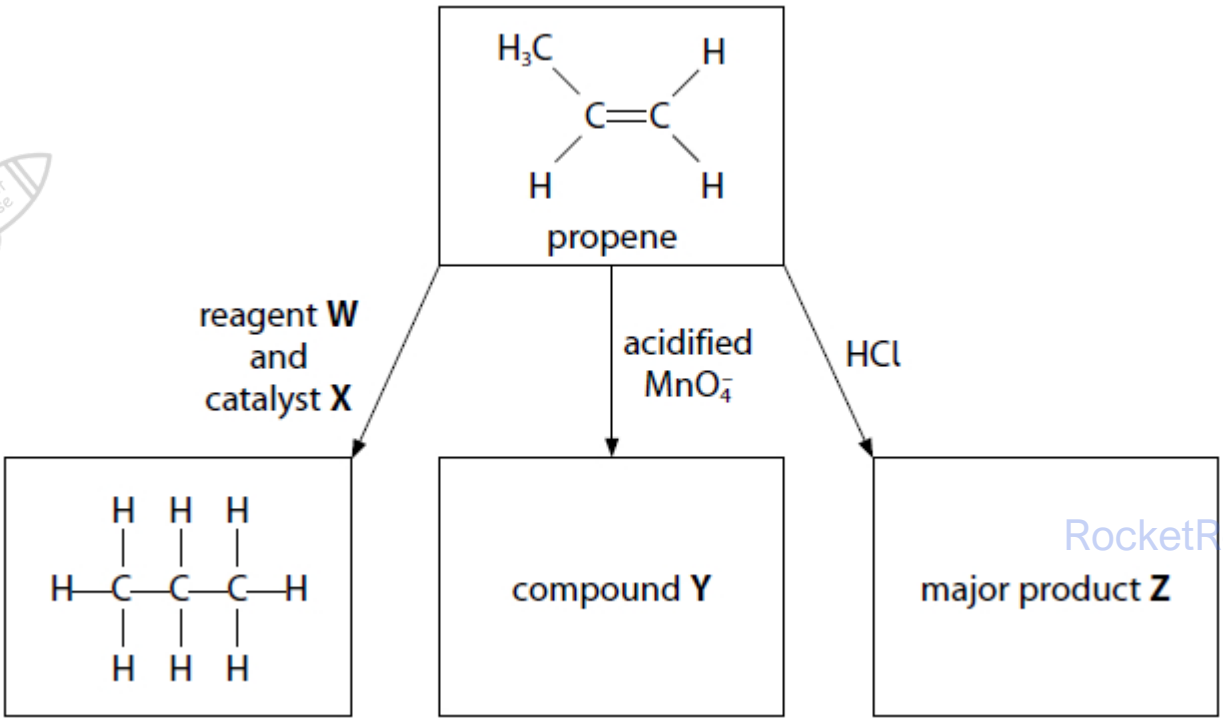
Q19.

Alkenes contain a double bond between two carbon atoms.

(a) Some reactions of propene are shown.

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(i) Give the names of reagent **W** and catalyst **X**.

(2)

Reagent **W**

.....

Catalyst **X**

.....

(ii) Draw the displayed formula of compound **Y**.

(1)

(iii) Draw the skeletal formula of the major product **Z**.

(1)

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(b) Ethene reacts with steam in the presence of a catalyst to form ethanol.

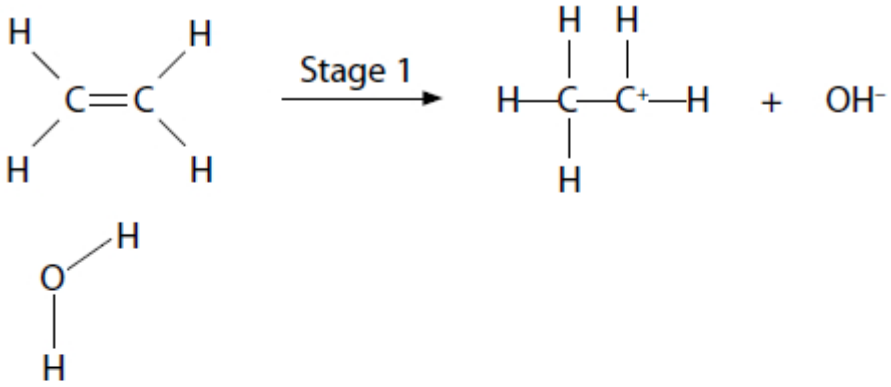
The mechanism takes place in two stages.

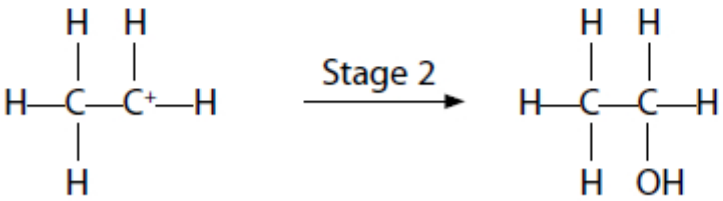
(i) Complete the simplified mechanism for the reaction by adding curly arrows and the relevant dipole.

(4)

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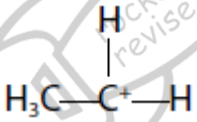
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:OH⁻

(ii) Predict the shape of the intermediate ion with reference to the positively-charged carbon. Justify your answer.



(3)

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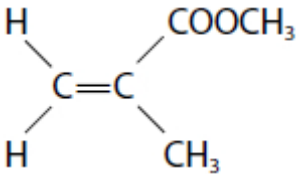
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(c) Methyl 2-methylpropenoate has the structure:



Draw a section of the polymer formed from methyl 2-methylpropenoate, showing two repeat units.

(2)

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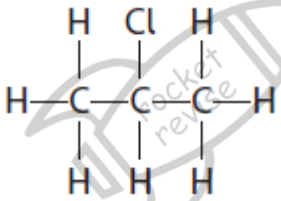
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(Total for question = 13 marks)

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Q20.

This question is about 2-Chloropropane.



(a) 2-Chloropropane has a relative molecular mass of 78.5 g mol^{-1}.

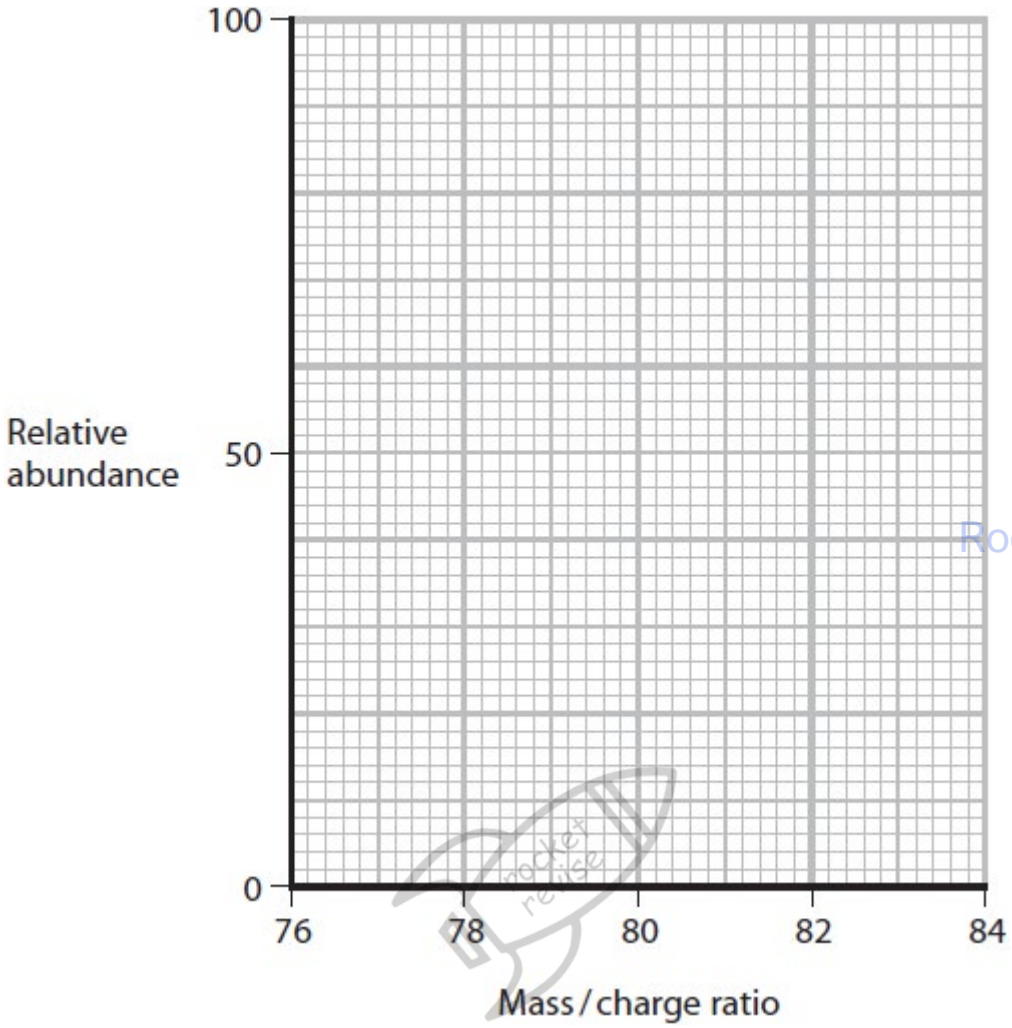
Chlorine has two common isotopes, ^{35}Cl and ^{37}Cl.
There are three times more ^{35}Cl atoms than ^{37}Cl atoms.
The main isotope of hydrogen is ^{1}H and that of carbon is ^{12}C.
The diagram shows a mass spectrum grid.
Draw the peaks for the molecular ions of 2-Chloropropane resulting from these isotopes.

(2)

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(b) 2-Chloropropane can be produced by reacting propane with chlorine in a homolytic free radical reaction.

$$CH_3CH_2CH_3 + Cl_2 \rightarrow CH_3CHClCH_3 + HCl$$

(i) Show the initiation step of this reaction.
Include appropriate arrows and the conditions necessary for this step.

(2)

.....

(ii) Using your answer to (b)(i), state what is meant by the terms homolytic and free radical.

(2)

homolytic

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free radical

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(iii) Suggest why this method has limited use in the synthesis of organic compounds.

(1)

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(c) 2-Chloropropane can also be produced from the reaction of propene with hydrogen chloride.

(i) Give the mechanism for this reaction.
Include curly arrows and relevant dipoles and lone pairs.

(4)

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(ii) Explain why only a small amount of 1-chloropropane is produced in this reaction.

(2)

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(Total for question = 13 marks)

Q21.

Both butter and margarine are fats used in cooking.

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(a) Margarines are sold as unsaturated fats.

State the meaning of the term unsaturated.

(1)

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.....

(b) (i) The degree of unsaturation can be determined by the reaction with bromine water.

Name the type of reaction.

(1)

.....

Data using 0.5 g of some unsaturated fat in this bromination is given in the table.

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Average number of C═C bonds per molecule	Volume of 0.0625 mol dm^{-3} bromine water / cm^3
1	28.3
2	57.3
3	86.9
4	115

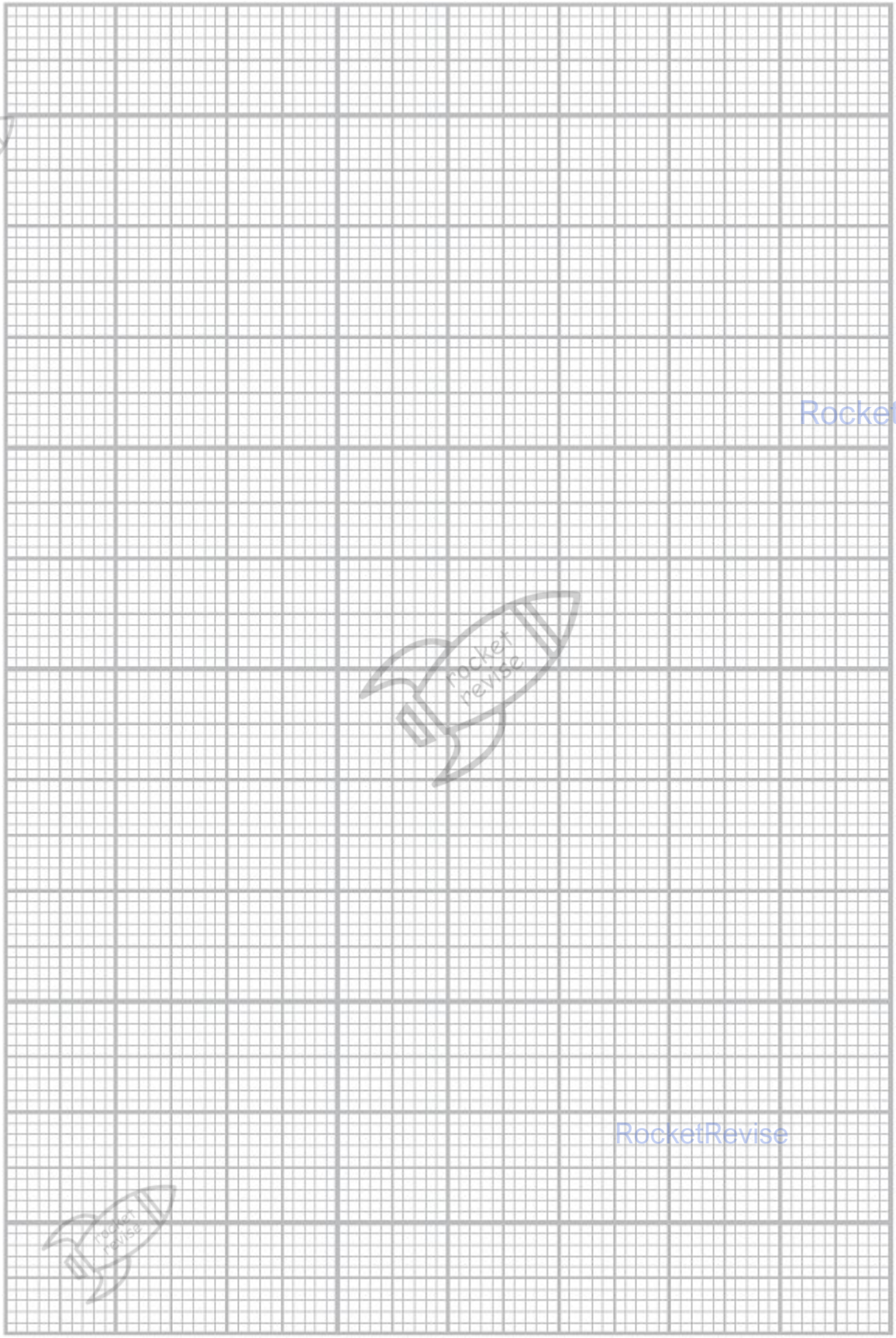
(ii) Plot a graph of the data. Include a line of best fit.

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(iii) Data for the bromination of 0.5 g samples of a margarine are shown.

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Trial	Volume of 0.0625 mol dm^{-3} bromine water / cm^3
1	36.9
2	34.1
3	39.3
4	32.5

Calculate the mean volume of bromine water using all results in the table.

(1)

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(iv) Determine the average number of C═C bonds per molecule of the unknown sample to 2 significant figures, using your graph.

(1)

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(c) (i) Using the simplest alkene as an example, draw the mechanism to show the reaction with liquid bromine.

(4)

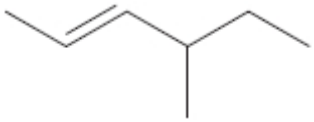
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(ii) Name this compound by applying IUPAC rules.

(1)

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(d) Explain the meaning of "trans" in "trans" fats, using a **skeletal** formula showing four carbon atoms in your answer.

(2)

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(Total for question = 14 marks)

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Q22.

Many synthetic polymers are used in the home.
Some are used as containers and others as coatings.

PVC and PTFE are two such examples.

(a) Draw the **displayed** formula of each of the monomers for these polymers.

(2)

chloroethene tetrafluoroethene

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(b) Incineration and recycling are two methods of disposing of polymer waste.

Give **one** advantage and **one** disadvantage for each method.

(4)

Incineration

advantage .....

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disadvantage.....

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Recycling

advantage .....

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disadvantage.....

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(Total for question = 6 marks)

Q23.

Compounds **A**, **B**, **C** and **D** all have the molecular formula C_4H_8.

A, **B** and **C** each contain one double bond, but **D** does not.

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A and **B** are geometric isomers of each other.

(a) Deduce a possible structure and name for each compound.

(4)

Possible structure of **A**

Name

.....

Possible structure of **B**

Name

.....

Possible structure of **C**

Name

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Possible structure of **D**

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Name

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(b) The carbon–carbon double bond consists of a σ bond and a π bond.

Describe the difference between the σ bond and the π bond.
Include a labelled diagram in your answer.

(4)

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(c) Give **two** reasons why compounds **A** and **B** exist as geometric isomers.

(2)

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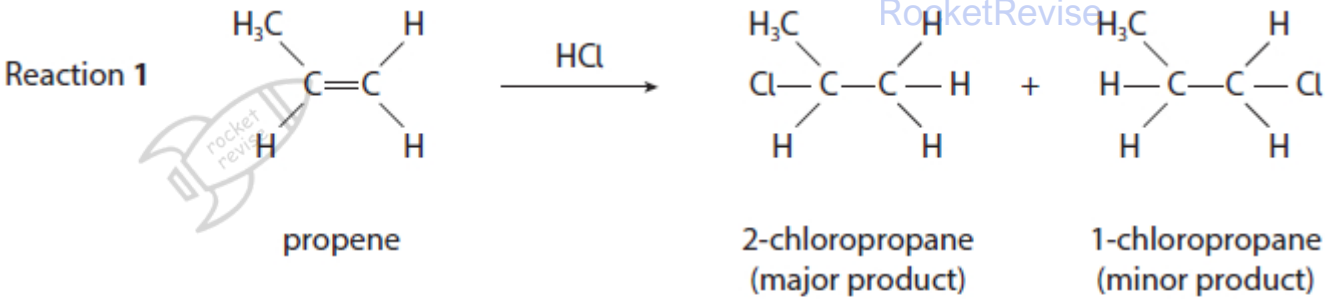
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(Total for question = 10 marks)

Q24.

This question is about two reactions of propene.



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(a) (i) Name the mechanism and type of reaction for Reaction **1**.

(1)

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(ii) Explain why the structures of the intermediates in Reaction **1** mean that 2‑chloropropane is formed in greater yield than 1‑chloropropane.

(3)

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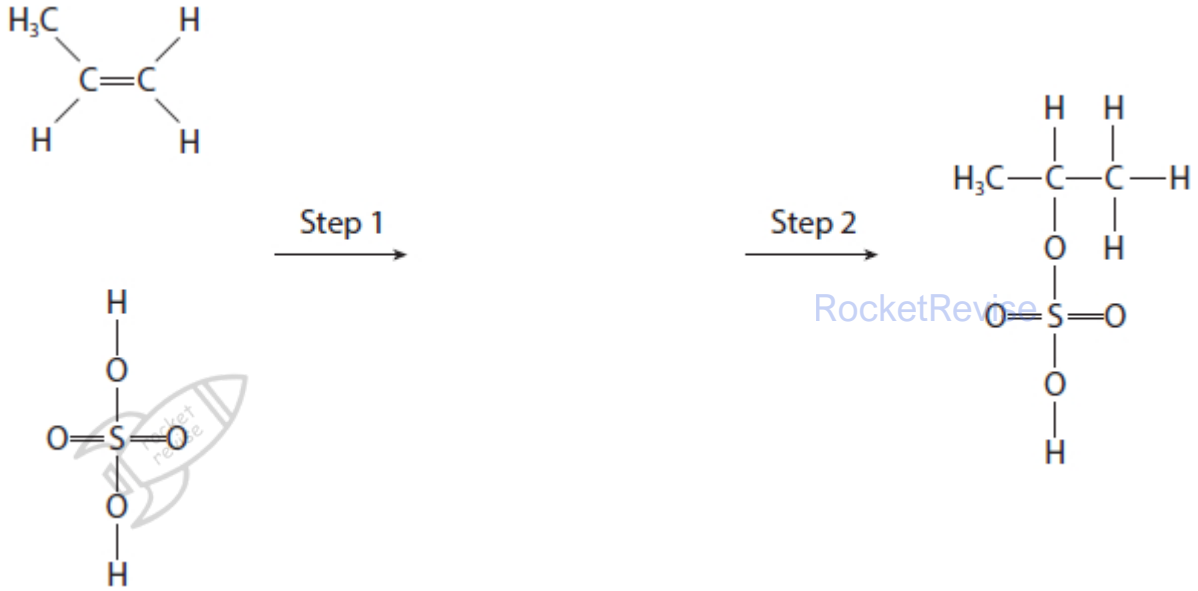
(b) Reaction **2** proceeds in two stages.

The equation for the first stage is shown.

$$CH_3CH{=}CH_2 + H_2SO_4 \rightarrow CH_3CH(OSO_2OH)CH_3$$

The mechanism for this reaction is the same as Reaction **1**.
Complete the mechanism by including the intermediate structure, curly arrows and relevant dipoles, charges and lone pairs.

(4)



(Total for question = 8 marks)

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Q25.

Chlorine reacts with both alkanes and alkenes.

(a) Three steps in the mechanism for the reaction of chlorine with butane to form 1-chlorobutane are shown.

Step 1 $Cl_2 \rightarrow 2Cl\bullet$

Step 2 $Cl\bullet + CH_3CH_2CH_2CH_3 \rightarrow CH_3CH_2CH_2CH_2\bullet + HCl$

Step 3 $CH_3CH_2CH_2CH_2\bullet + Cl_2 \rightarrow CH_3CH_2CH_2CH_2Cl + Cl\bullet$

(i) Name the mechanism and type of this reaction.

(1)

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(ii) Give the essential condition for Step **1** to occur.

(1)

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(iii) Name the type of step illustrated by Step **3**.

(1)

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(iv) A different type of step, other than the ones shown, also occurs in the reaction.
Describe this type of step, including two possible examples which might occur in this reaction.

(4)

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(b) Chlorine is not a polar molecule but can react with alkenes through an electrophilic addition reaction.

(i) State what is meant by the term polar molecule and why chlorine is not polar.

(2)

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(ii) Explain why chlorine reacts with alkenes even though it is not a polar molecule.

(2)

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(Total for question = 11 marks)

Q26.

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A sample of a volatile liquid **X** was vaporised.
At 473 K and 110 000 Pa, 3.50 g of **X** occupies 1.79 dm^3.

(a) (i) Calculate the number of moles of **X** in the sample.

[R = 8.31 J mol^{-1} K^{-1}]
Use $pV = nRT$

(4)

(ii) Deduce the relative molecular mass of **X**.

(1)

(iii) Compound **X** contains 85.7 % carbon and 14.3 % hydrogen by mass.
Calculate the empirical formula of **X**.

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(2)

(iv) Deduce the molecular formula of **X**.

(1)

(b) A few drops of a purple solution of acidified potassium manganate(VII) were added to a sample of liquid X and shaken.

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The solution remained purple.
State what can be deduced about the structure of **X** from this experiment.

(1)

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(c) Identify, by name or formula, **two** possible structural isomers of **X** using your answers to (a)(iv) and (b).

(2)

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(Total for question = 11 marks)

Q27.

Chloroethene and 1-chloropropene are the first two members of a homologous series. 1-Chloropropene exists as two geometric isomers.

(a) (i) Draw the **displayed** formulae of chloroethene and one geometric isomer of 1-chloropropene.

(2)

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chloroethene 1-chloropropene

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(ii) Explain why chloroethene and 1-chloropropene are members of a homologous series.

(2)

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(iii) Explain why 1-chloropropene exists as two geometric isomers, but chloroethene does not.

(3)

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(b) Chloroethene and 1-chloropropene are both monomers.

(i) Draw the structure of poly(1-chloropropene).

(2)

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(ii) The polymer of chloroethene, known as PVC, is used to make water pipes and gutters, which were previously made of metal.

Suggest **two** advantages of using PVC instead of metal in water pipes and gutters, other than cost.

(2)

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(Total for question = 11 marks)

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